All-Addition Hyperspectral Compressed Sensing for Metasurface-Driven Miniaturized Satellite

Chia-Hsiang Lin, *Member, IEEE*, and Tzu-Hsuan Lin, *Graduate Student Member, IEEE*

***Abstract*— Hyperspectral compressed sensing (HCS) for the miniaturized satellite is challenging mainly due to the required lightweight onboard hardware and the necessary low sampling rate. Optical devises involved in conventional HCS include spectral splitter (SS), digital micromirror device (DMD) array, and cylindrical lens (CL), while DMD array (used for implementing random projections) is known to be bulky. We are, hence, thinking of the possibility of just using SS and CL, mathematically meaning that only the deterministic addition operator is available in the coding stage. Another recent advance in the multifunctional metamaterial also motivates us to solve such mathematical challenges. Specifically, SS and CL can be designed on a single nanoscale metasurface (flat), more in line with the miniaturized satellite application. We show that this all-addition coding scheme is achievable though it induces a far more challenging decoding stage, for which a convex decoding criterion is proposed based on self-similarity, a well-known property in imaging inverse problems. Self-similarity regularizer has recently been explicitly defined as a convex function and is demonstrated to be effective in decoding even with a low sampling rate. In addition, as the fundamental role of the John ellipsoid (JE) has been revealed in recent hyperspectral analysis literature, we build a JE-based convex analysis framework to ensure exact recovery even with low data purity. Experimental evidence shows the superiority of the proposed method.**

***Index Terms*— Convex optimization, hyperspectral compressed sensing (HCS), miniaturized satellite, self-similarity.**

I. Introduction

COMPRESSED sensing enables highly efficient computational imaging, for which metamaterial starts to play an essential role in aperture design [1]–[5]. Metasurface technique has drawn considerable attention in very recent remote sensing and communication communities, such as intelligent reflecting surfaces (IRSs) in 6G wireless systems [6], [7]. In particular, to match the trend of the miniaturized satellite, the metamaterial is also a good choice for designing lightweight onboard hardware. On the other hand, for satellite remote sensing, hyperspectral imagery has outstanding capability in material identification due to its densely populated spectral bands and the hyperspectral unmixing (HU) technology, having various applications, such as planetary exploration, environmental monitoring, and mineral identification and quantification [8]. Specifically, typical hyperspectral remote sensors record the electromagnetic scattering patterns of the underlying materials presented in the scene of interest over hundreds of narrowly spaced (typically, 5–10 nm) wavelengths continuously ranging from visible to near-infrared bands [9], [10]. Due to the huge volume of typical hyperspectral imagery, a low sampling rate is desired for efficient satellite communications [11]. In this work, we focus on developing hyperspectral compressed sensing (HCS) technique for miniaturized satellites.

Optical devices involved in conventional HCS hardware include spectral splitter (SS), digital micromirror device (DMD) array, and cylindrical lens (CL) [12]. Since the DMD array [used for implementing random projections (RPs)] is known to be easily aging and bulky, we are thinking of the possibility of just using SS and CL, mathematically meaning that only a deterministic addition operator is available in the coding stage. Developing such an all-addition HCS technique is also motivated by the recent advances in the so-called *multifunctional* metamaterial. More specifically, both functions of SS and CL can be compactly realized on a single flat metasurface, as demonstrated in the recent nanophotonics literature [13]–[15], quite appealing to the requirement of lightweight onboard hardware for miniaturized satellite applications. All these recent advances motivate us to solve the mathematical challenge: *Can we design an HCS encoder involving just the addition operator (without RP) and accordingly an effective decoder working well even at a very low sampling rate (e.g., 1%)?* The answer is affirmative. We will show that this all-addition coding scheme is achievable though it induces a far more challenging decoding stage.

Unlike the total-variation decoding scheme popularly adopted in several benchmark HCS methods, including hyperspectral coded aperture (HYCA) [16] and locally similar sparsity-based HU compressive sensing (LSSHUCS) [17], our encoder will be developed based on self-similarity, a strong prior underlying a series of impressive state-of-the-art contributions to imaging inverse problems [18]–[20]. Self-similarity is widely observed in natural images, including multispectral images, hyperspectral images (HSIs), videos, and magnetic resonance imaging sequences, and asserts that a real-world image is very likely to exhibit highly structured configurations containing similar patches at different locations [21]. Though self-similarity has been employed in several state-of-the-art

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works (e.g., multiresolution satellite imaging analysis [20]), it has never been explicitly defined until very recently [22]. In [22], for the first time, the self-similarity phenomenon has been explicitly defined as a convex function and is, hence, suitable for compressed sensing (CS) applications. Note that the convexity of the decoder has not only yielded great success in CS, but it is also critical in guaranteeing the convergence of the reconstruction algorithm [20], [22]. We will show that such a convex self-similarity function is effective in the decoding stage even with a low sampling rate. In addition, as the fundamental role of the John ellipsoid (JE) topology in the hyperspectral analysis has been revealed in recent mathematics and machine learning literature [10], [23], we build a JE-based convex analysis framework to ensure exact recovery under a very mild sufficient condition on hyperspectral data purity.

A. Related HCS Works

Next, we review the encoding/decoding schemes of some state-of-the-art HCS methods. To this end, we recall that each M-band pixel $\boldsymbol{x}_i \in \mathbb{R}^M$ in a HSI $\boldsymbol{X} = [\boldsymbol{x}_1, \ldots, \boldsymbol{x}_L]$ can be modeled as a convex combination of the involved material signatures, i.e., $\boldsymbol{x}_i = \boldsymbol{A}\boldsymbol{s}_i$ with abundances $\boldsymbol{s}_i$ satisfying $\boldsymbol{s}_i \geq \boldsymbol{0}$, $\boldsymbol{1}^T\boldsymbol{s}_i = 1$, where the nth column of $\boldsymbol{A}$ is the nth material signature vector [24]; by defining the abundance matrix as $\boldsymbol{S} = [\boldsymbol{s}_1, \ldots, \boldsymbol{s}_L]$, one can then concisely write $\boldsymbol{X} = \boldsymbol{A}\boldsymbol{S}$. Note that $\boldsymbol{A}$ is usually assumed to have affinely independent columns [24].

Let us begin by reviewing the RP method HYCA [16], where each hyperspectral pixel $\boldsymbol{x}_i \in \mathbb{R}^M$ is encoded as $\boldsymbol{A}_i\boldsymbol{x}_i$ with a random sampling matrix $\boldsymbol{A}_i \in \mathbb{R}^{q\times M}$ ($q \ll M$). This typical sensing strategy motivates the definition of affine independence property (AIP), meaning that a sensing strategy preserves the affine independence of signatures $\boldsymbol{A}$ in the HCS measurement:

Definition 1: Assume that the sampling matrix associated with the ith hyperspectral pixel $\boldsymbol{x}_i = \boldsymbol{A}\boldsymbol{s}_i$ (i.e., convex combination of signature vectors in $\boldsymbol{A}$) is denoted by $\boldsymbol{A}_i$. If $\boldsymbol{A}_i\boldsymbol{A}$ has affinely independent columns, $\forall i = i = 1, \ldots, L$, then the corresponding measurement strategy is said to satisfy AIP. □

A direct consequence is given in the following lemma, whose proof simply follows from the abundance nonnegativity $\boldsymbol{s}_i \geq \boldsymbol{0}$ and full-additivity $\boldsymbol{1}^T\boldsymbol{s}_i = 1$:

Lemma 1: If a measurement strategy satisfies AIP, then all the $\boldsymbol{s}_i$'s (and, hence, $\boldsymbol{x}_i$) can be recovered from $\boldsymbol{A}$ and its corresponding HCS measurement (i.e., $\boldsymbol{A}_i\boldsymbol{x}_i$). □

To alleviate the complexity of the recovery algorithm, HYCA assumes that the sampling matrix $\boldsymbol{A}_i$ is drawn from a finite pool, i.e., $\boldsymbol{A}_i \in \{\boldsymbol{H}_1, \ldots, \boldsymbol{H}_{n_h}\}$. Also, the image is divided into nonoverlapping windows of size $w \times w$ (with $w^2 = n_h$). Then, pixels in a given window are one-to-one associated with $\boldsymbol{H}_1, \ldots, \boldsymbol{H}_{n_h}$, and such association patterns are the same across all windows. The decoder of HYCA adopts the aforementioned linear mixing model $\boldsymbol{X} = \boldsymbol{A}\boldsymbol{S}$, where the abundance matrix $\boldsymbol{S} \geq \boldsymbol{0}$ is nonnegative by nature, and $\boldsymbol{A}$ is assumed known. By defining the vectorizations $\boldsymbol{x} = \text{vec}(\boldsymbol{X})$ and $\boldsymbol{s} = \text{vec}(\boldsymbol{S})$, the HCS measurement can be concisely written as $\boldsymbol{y} = \text{bdiag}(\boldsymbol{A}_1, \ldots, \boldsymbol{A}_L)\boldsymbol{x} = \text{bdiag}(\boldsymbol{A}_1\boldsymbol{A}, \ldots, \boldsymbol{A}_L\boldsymbol{A})\boldsymbol{s} \triangleq \boldsymbol{K}\boldsymbol{s}$, where "bdiag" is the block-diagonal operator. In the decoding stage, HYCA solves a convex optimization problem

$$\min_{\boldsymbol{s}\geq\boldsymbol{0}} \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{K}\boldsymbol{s}\|_2^2 + \lambda \text{TV}(\boldsymbol{s})$$

where "TV" denotes the total-variation regularization with strength controlled by $\lambda \geq 0$.

Another state-of-the-art HCS work employing convex TV-decoder is LSSHUCS [17], which assumes the availability of a redundant library $\boldsymbol{W}$ [collecting r material signatures (endmembers)] and models the hyperspectral data as $\boldsymbol{X} = \boldsymbol{W}\boldsymbol{H}$. It is obvious that, due to the redundancy, $\boldsymbol{H}$ is a row-sparse matrix. An alternative when such library is not readily available is to compute $\boldsymbol{W}$ via some endmember extraction algorithm [24]. LSSHUCS also performs HCS along the spectral dimension; specifically, a Gaussian random sampling matrix $\boldsymbol{G}$ is adopted to obtain the measurement $\boldsymbol{F} = \boldsymbol{G}\boldsymbol{X} \in \mathbb{R}^{m_a\times L}$. The decoder makes a strong assumption that each pixel must be some column of $\boldsymbol{W}$, implying a highly sparse pattern of $\boldsymbol{H} \in \{0, 1\}^{r\times L}$, which, together with the row-sparsity nature of $\boldsymbol{H}$ helping automatic endmember selection, gives rise of the first objective function $g_{\text{spectral}}(\boldsymbol{H}) = \|\boldsymbol{H}\|_1$. On the other hand, the locally similar sparsity (meaning a spatially smooth $\boldsymbol{H}$) motivates the second objective function $g_{\text{spatial}}(\boldsymbol{H}) = \text{TV}(\boldsymbol{H})$, leading to the convex decoding criterion

$$\min_{\boldsymbol{H}\geq\boldsymbol{0},\ \boldsymbol{G}\boldsymbol{W}\boldsymbol{H}=\boldsymbol{F}} g_{\text{spectral}}(\boldsymbol{H}) + \lambda_{\text{LSSHUCS}}\, g_{\text{spatial}}(\boldsymbol{H})$$

where $\lambda_{\text{LSSHUCS}} > 0$ balances the two objective functions.

Unlike HYCA, the spectral compressive acquisition (SpeCA) [12] method does not assume the availability of the subspace and learns the data subspace directly from the RPs. The RP measurements consist of two parts, $\boldsymbol{Y}_a = \boldsymbol{G}'\boldsymbol{X} \in \mathbb{R}^{m_a\times L}$ and $\boldsymbol{Y}_b = [\boldsymbol{B}_1\boldsymbol{x}_{i_1}, \ldots, \boldsymbol{B}_v\boldsymbol{x}_{i_v}] \in \mathbb{R}^{m_b\times v}$, where the measurement matrices $\boldsymbol{G}', \boldsymbol{B}_1, \ldots, \boldsymbol{B}_v$ are independently drawn at random from Gaussian/Rademacher/Bernoulli distributions. SpeCA still employs the low-rank model $\boldsymbol{X} = \boldsymbol{A}\boldsymbol{S} = \boldsymbol{A}[\boldsymbol{s}_1, \ldots, \boldsymbol{s}_L]$, leading to $\boldsymbol{Y}_a = \boldsymbol{F}\boldsymbol{S}$ with $\boldsymbol{F} \triangleq \boldsymbol{G}'\boldsymbol{A}$. Then, matrix factorization is performed via singular value decomposition (SVD) or HU to obtain $\boldsymbol{F}$, which is assumed to be full-column rank and used to recover $\widehat{\boldsymbol{S}} = (\boldsymbol{F}^T\boldsymbol{F})^{-1}\boldsymbol{F}^T\boldsymbol{Y}_a$. Finally, the kth column of $\boldsymbol{Y}_b$ is $\boldsymbol{y}_{b_k} = \boldsymbol{B}_k\boldsymbol{x}_{i_k} = \boldsymbol{B}_k\boldsymbol{A}\boldsymbol{s}_{i_k}$, from which one can derive $\text{vec}(\boldsymbol{Y}_b) = \boldsymbol{D}'\text{vec}(\boldsymbol{A})$ with $\boldsymbol{D}' \triangleq [(\widehat{\boldsymbol{s}}_{i_1}\otimes\boldsymbol{B}_1^T), \ldots, (\widehat{\boldsymbol{s}}_{i_v}\otimes\boldsymbol{B}_v^T)]^T$ assumed to be full-column rank. Thus, $\text{vec}(\boldsymbol{A}) = (\boldsymbol{D}'^T\boldsymbol{D}')^{-1}\boldsymbol{D}'^T\text{vec}(\boldsymbol{Y}_b)$ is used to obtain $\widehat{\boldsymbol{A}}$ and the reconstructed HSI $\widehat{\boldsymbol{X}} = \widehat{\boldsymbol{A}}\widehat{\boldsymbol{S}}$. Simply speaking, SpeCA reconstructs the spatial and spectral information, respectively, from the two-part RP measurements.

Finally, joint tensor reconstruction via 3-D total variation (JTenRe3DTV) [25] considers the HSI as a three-way tensor $\mathcal{X} \in \mathbb{R}^{L_1\times L_2\times M}$ ($L_1L_2 = L$) and applies the randomly permuted Hadamard transform $\boldsymbol{\Phi}' = \boldsymbol{D}'' \cdot \boldsymbol{H}' \cdot \boldsymbol{P}$ to the HSI in the encoding stage to obtain the measurement $\boldsymbol{y} = \boldsymbol{\Phi}'(\mathcal{X})$, where $\boldsymbol{D}''$ (respectively, $\boldsymbol{P}$) performs random downsampling (respectively, random permutation) and $\boldsymbol{H}'$ is the Walsh-Hadamard transform. It is obvious that the mode-3 matricization of $\mathcal{X}$ is low-rank (hence, compressible)

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(i.e., $\mathcal{X}_{(3)} = \boldsymbol{X} = \boldsymbol{AS}$), where mode-$n$ matricization of a tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ is denoted as $\mathcal{X}_{(n)}$ with $[\mathcal{X}]_{i_1,\ldots,i_N}$ mapped to $[\mathcal{X}_{(n)}]_{i_n,j}$ ($j \triangleq 1 + \sum_{k=1,k\neq n}^{N}(i_k - 1)J_k$ with $J_k \triangleq \prod_{m=1,m\neq n}^{k-1} I_m$). Assuming that the matrix unfoldings in the two spatial modes $\mathcal{X}_{(1)}$ and $\mathcal{X}_{(2)}$ are also compressible, one has the low-rank Tucker decomposition $\mathcal{X} \approx \mathcal{C} \times_1 \boldsymbol{U}_1 \times_2 \boldsymbol{U}_2 \times_3 \boldsymbol{U}_3$ based on the global spatial–spectral correlation of typical HSI. Here, the mode-n multiplication of a tensor $\mathcal{C}$ with a matrix $\boldsymbol{U}$, denoted by $\mathcal{C} \times_n \boldsymbol{U}$, is to left-multiply $\boldsymbol{U}$ to each vector along dimension-n in $\mathcal{C}$; to be precise, $[\mathcal{C} \times_n \boldsymbol{U}]_{i_1,\ldots,i_{n-1},j,i_{n+1},\ldots,i_N} \triangleq \sum_m \boldsymbol{U}_{j,m}[\mathcal{C}]_{i_1,\ldots,i_{n-1},m,i_{n+1},\ldots,i_N}$. The decoding criterion is, hence, designed as

$$\min_{\mathcal{X},\mathcal{E},\mathcal{C},\boldsymbol{U}_i} \frac{1}{2}\|\mathcal{E}\|_F^2 + \lambda\|\mathcal{X}\|_{\text{3D-TV}} \quad \text{s.t. } \boldsymbol{y} = \boldsymbol{\Phi}'(\mathcal{X}), \quad \mathcal{X} = \mathcal{C} \times_1 \boldsymbol{U}_1 \times_2 \boldsymbol{U}_2 \times_3 \boldsymbol{U}_3 + \mathcal{E}$$

where $\lambda > 0$, and $\|\cdot\|_F$ and $\|\cdot\|_{\text{3D-TV}}$ are the Frobenius norm and the 3-D TV regularizer, respectively. JTenRe3DTV adopts a holistic compressive manner, i.e., requiring the entire HSI cube, which is not suitable for miniaturized satellite application.

B. Notation and Article Organization

Some standard notations/definitions used throughout this article are collectively summarized hereinafter. A vector is called k-sparse if it has maximally k nonzero entries. $\mathbb{R}^n$ is the n-dimensional Euclidean space, and $\mathbb{R}^{m\times n}$ is the set of all the $m \times n$ real-valued matrices. $\mathbb{Z}_{++}$ is the set of positive integers. $\mathcal{I}_Z \triangleq \{1, \ldots, Z\}$ for any given positive integer Z. $\mathbf{1}$ is the all-one vector, and $\mathbf{0}$ is a zero vector or matrix with dimension specified in the subscript. $\boldsymbol{e}_i^{(n)}$ is the ith unit vector of $\mathbb{R}^n$. $[\boldsymbol{v}]_{a:b}$ is the subvector of a given vector $\boldsymbol{v}$ formed by its ath to bth entries. sup S denotes supremum of a set S [26]. $\|\cdot\|_F$ and $\|\cdot\|_p$ denote the Frobenius norm and p-norm, respectively. vec($\boldsymbol{M}$) denotes the vectorization of the matrix $\boldsymbol{M}$, while $\text{vec}^{-1}_{m\times n}(\boldsymbol{v})$ for a given vector $\boldsymbol{v}$ is the $m\times n$ matrix $\boldsymbol{M}$ satisfying vec($\boldsymbol{M}$) = $\boldsymbol{v}$. "conv" denotes the convex hull [26]. $\otimes$ denotes the Kronecker product. Finally, bdiag($\boldsymbol{M}_1, \ldots, \boldsymbol{M}_K$) is the block diagonal matrix with the ith diagonal block given by $\boldsymbol{M}_i$.

The remaining parts of this article are organized as follows. In Section II, we present the proposed HCS method, as well as the related analysis, including convergence, complexity, and identifiability. Experimental results are summarized and discussed in Section III, where a comparison with state-of-the-art peer methods can also be found. Finally, conclusions are drawn in Section IV, while the mathematical details are collectively presented in the Appendix.

II. Proposed HCS Method

In CS, the compressed acquisition can be written as

$$\boldsymbol{y} = \boldsymbol{\Phi}\boldsymbol{x} \tag{1}$$

where $\boldsymbol{\Phi} \in \mathbb{R}^{m\times n}$ is the measurement matrix with $m \ll n$, while the decoder aims to solve the above underdetermined linear system for recovering the signal $\boldsymbol{x}$. Though the solution is not unique in general, $\boldsymbol{x}$ would still be uniquely recovered under some sparsity conditions. To get a sense, the compressed acquisition $\boldsymbol{y} = \mathbf{1}_2$ under the measurement matrix $\boldsymbol{\Phi} = [\boldsymbol{e}_1^{(2)}, \boldsymbol{e}_2^{(2)}, \mathbf{1}_2]$ corresponds to multiple possible causal factors, but, when $\boldsymbol{x}$ is constrained to be k-sparse with $k = 1$, the only possible causal factor is then uniquely given by $\boldsymbol{x}^\star = \boldsymbol{e}_3^{(3)}$. However, the signal $\boldsymbol{x}$ may not always be sparse by nature. Thus, another key stage in CS is the design of sparsifying matrix or basis matrix $\boldsymbol{\Psi} \in \mathbb{R}^{n\times p}$, which represents the signal by a set of sparse coefficients $\boldsymbol{s}$, i.e.,

$$\boldsymbol{x} = \boldsymbol{\Psi}\boldsymbol{s}. \tag{2}$$

With this transform, the sparsity is again applicable with respect to the linear system $\boldsymbol{\Theta} = \boldsymbol{\Phi}\boldsymbol{\Psi}$; specifically, the sparse coefficients $\boldsymbol{s}^\star$ would be obtained by solving $\boldsymbol{y} = \boldsymbol{\Theta}\boldsymbol{s}$, followed by recovering the signal as $\boldsymbol{x}^\star = \boldsymbol{\Psi}\boldsymbol{s}^\star$. Nevertheless, in practice, sparsity itself is not sufficient for effective decoding; that is why more sophisticated regularization schemes (e.g., sparsity in the transformed gradient domain, i.e., TV) are extensively investigated. In this work, based on the nature of HSI, self-similarity will be adopted as the regularizer. To summarize, the backbone of CS includes measurement matrix design, basis matrix design, and decoding criterion design, which will be developed in Sections II-B, II-C and II-D, respectively. Preliminary and analysis are presented in Sections II-A and II-E, respectively.

A. Preliminary

Some required basics are collectively and concisely presented in this section. We begin by talking about restricted isometry property (RIP) [27], followed by the hyperspectral data purity γ [28], [29] and JE in blind hyperspectral signal processing [10].

Definition 2: Let $\boldsymbol{\Sigma}_k$ be the set of all k-sparse vectors. If the measurement matrix $\boldsymbol{\Phi}$ satisfies

$$(1-\delta_k)\|\boldsymbol{x}\|_2^2 \le \|\boldsymbol{\Phi}\boldsymbol{x}\|_2^2 \le (1+\delta_k)\|\boldsymbol{x}\|_2^2 \quad \forall \boldsymbol{x} \in \boldsymbol{\Sigma}_k$$

for some $\delta_k \in (0, 1)$, then $\boldsymbol{\Phi}$ is said to satisfy order-k RIP. □

The RIP condition is basically saying that, for any $\widetilde{\boldsymbol{\Phi}}$ formed by k columns in $\boldsymbol{\Phi}$, the matrix $\widetilde{\boldsymbol{\Phi}}^T\widetilde{\boldsymbol{\Phi}}$ is positive definite (PD) with eigenvalues in $[1-\delta_k, 1+\delta_k]$, and it ensures exact sparse recovery as implied by the following lemma [27].

Lemma 2: Let $\boldsymbol{\Phi} \in \mathbb{R}^{m\times n}$ satisfy the RIP of order $2k$ with $\delta_{2k} < \sqrt{2} - 1$. Let $\boldsymbol{x} \in \mathbb{R}^n$ (with measurement $\boldsymbol{y} = \boldsymbol{\Phi}\boldsymbol{x}$), and let $\widehat{\boldsymbol{x}}$ be a solution of the associated basis pursuit (BP) problem, i.e., $\widehat{\boldsymbol{x}} = \arg\min_{\boldsymbol{y}=\boldsymbol{\Phi}\widetilde{\boldsymbol{x}}} \|\widetilde{\boldsymbol{x}}\|_1$. Then

$$\|\boldsymbol{x} - \widehat{\boldsymbol{x}}\|_2 \le c\left(\frac{\sigma_k(\boldsymbol{x})_1}{\sqrt{k}}\right)$$

where c is a constant, and $\sigma_k(\boldsymbol{x})_1 \triangleq \min_{\widetilde{\boldsymbol{x}}\in\boldsymbol{\Sigma}_k} \|\boldsymbol{x} - \widetilde{\boldsymbol{x}}\|_1$ is the so-called best k-term approximation error. □

Remarkably, if $\boldsymbol{x}$ is k-sparse (implying $\sigma_k(\boldsymbol{x})_1 = 0$), the BP recovery $\widehat{\boldsymbol{x}} = \boldsymbol{x}$ is exact.

Next, we briefly recall the concept of data purity γ of a given hyperspectral data $\boldsymbol{X} = \boldsymbol{AS} = \boldsymbol{A}[\boldsymbol{s}_1, \ldots, \boldsymbol{s}_L]$ [28].

Definition 3: The data purity level γ of a given hyperspectral data $\boldsymbol{X}$ is defined as

$$\gamma \triangleq \sup\{r \le 1 \mid (r\mathcal{B}_N) \cap \mathcal{T}_e \subseteq \text{conv}\{\boldsymbol{S}\}\} \tag{3}$$

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where $\text{conv}\{\boldsymbol{S}\} \triangleq \text{conv}\{\boldsymbol{s}_1, \ldots, \boldsymbol{s}_L\} \subseteq \mathbb{R}^N$ with $\boldsymbol{s}_i$ being the abundance vector of the ith pixel, and $\mathcal{B}_N \triangleq \{\boldsymbol{x} \in \mathbb{R}^N \mid \|\boldsymbol{x}\|_2 \leq 1\}$ and $\mathcal{T}_e \triangleq \text{conv}\{\boldsymbol{e}_1^{(N)}, \ldots, \boldsymbol{e}_N^{(N)}\} \subseteq \mathbb{R}^N$ are the unit ball and the unit simplex, respectively. □

Interested readers may refer to [28] for a comprehensive discussion about the physical meaning of γ. Simply speaking, one can show that its upper bound is $\gamma = 1$, corresponding to the well-known pure-pixel assumption (PPA) [30] in hyperspectral remote sensing (i.e., $\boldsymbol{e}_i^{(N)} \in \{\boldsymbol{s}_1, \ldots, \boldsymbol{s}_L\}\ \forall i \in \mathcal{I}_N$) [28], which is the purest case. By contrast, the most heavily mixed scenario $\boldsymbol{s}_i = [1/N, \ldots, 1/N]\ \forall i \in \mathcal{I}_L$ (i.e., the proportions of the N materials in each pixel are all $1/N$) corresponds to its lower bound $\gamma = 1/\sqrt{N}$ [28]. All in all, $\gamma \in [1/\sqrt{N}, 1]$, and a larger γ indicates a higher data purity [28]. Remarkably, our exact recovery guarantee does not require the PPA $\gamma = 1$ even though PPA is required almost in every well-known hyperspectral analysis frameworks [30]; instead, we only need a mild condition $\gamma > 1/\sqrt{N-1}$, as will be seen soon.

Now, we talk about JE, whose fundamental role in hyperspectral analysis has been revealed in recent mathematics and machine learning literature [10], [23]. Given a set of data points (i.e., hyperspectral pixel vectors), JE is the maximum-volume ellipsoid inscribed in the data convex hull. A $(N-1)$-dimensional ellipsoid $\mathcal{E}$ in $\mathbb{R}^M$ can be parameterized by a vector $\boldsymbol{c} \in \mathbb{R}^M$ and some full column-rank $\boldsymbol{F} \in \mathbb{R}^{M\times(N-1)}$, i.e.,

$$\mathcal{E}(\boldsymbol{F}, \boldsymbol{c}) = \{\boldsymbol{F}\boldsymbol{\alpha} + \boldsymbol{c} \mid \|\boldsymbol{\alpha}\|_2 \leq 1\}. \tag{4}$$

From the aforementioned equation $\boldsymbol{X} = \boldsymbol{A}\boldsymbol{S}$, as well as some widely adopted assumptions [10] in hyperspectral analysis, i.e., the following holds.

A1: $\boldsymbol{s}_i \geq \boldsymbol{0}_N$, $\boldsymbol{1}_N^T \boldsymbol{s}_i = 1$.
A2: $\boldsymbol{A} \in \mathbb{R}^{M\times N}$ has full column rank with $N \geq 3$.
A3: $\boldsymbol{S} \in \mathbb{R}^{N\times L}$ has full row rank.

It is not difficult to see that the dimension of the data convex hull $\text{conv}\{\boldsymbol{X}\} = \text{conv}\{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_L\}$ is $N-1$. Therefore, one can assume without loss of generality (w.l.o.g.) that the JE is in the dimension-reduced (DR) space $\mathbb{R}^M = \mathbb{R}^{N-1}$, meaning that $\boldsymbol{F} \in \mathbb{R}^{M\times(N-1)}$ is a square matrix; $\boldsymbol{F}$ can be further assumed w.l.o.g. to be PD since $\mathcal{E}(\boldsymbol{F}, \boldsymbol{c}) = \mathcal{E}(\sqrt{\boldsymbol{F}\boldsymbol{F}^T}, \boldsymbol{c})$ [26]. With the $\mathcal{H}$-polytope representation of data convex hull $\text{conv}\{\boldsymbol{X}\} = \bigcap_{i=1}^m \{\boldsymbol{x} \mid \boldsymbol{b}_i^T \boldsymbol{x} \leq h_i\}$, where the half-space parameter $\{(\boldsymbol{b}_i, h_i)\}_{i=1}^m$ can be easily obtained by calling the off-the-shelf QuickHull solver [31], one can see that JE problem is nothing but a convex conic program [26], i.e.,

$$\begin{aligned}(\boldsymbol{F}^\star, \boldsymbol{c}^\star) \in \arg \min_{\boldsymbol{F}\in\mathbb{S}_{++}^{N-1},\ \boldsymbol{c}\in\mathbb{R}^{N-1}} \quad & \log\det(\boldsymbol{F}^{-1}) \\ \text{s.t.} \quad & \|\boldsymbol{F}\boldsymbol{b}_i\| \leq h_i - \boldsymbol{b}_i^T \boldsymbol{c} \quad \forall i \in \mathcal{I}_m\end{aligned} \tag{5}$$

where "$\|\boldsymbol{F}\boldsymbol{b}_i\| \leq h_i - \boldsymbol{b}_i^T \boldsymbol{c}\ \forall i \in \mathcal{I}_m$" are second-order cone constraints provably equivalent to the single data-inscribing constraint "$\mathcal{E}(\boldsymbol{F}, \boldsymbol{c}) \subseteq \text{conv}\{\boldsymbol{X}\}$" (i.e., the ellipsoid should be constrained to be inscribed in the data convex hull), while the objective function $\log\det(\boldsymbol{F}^{-1})$ (inversely proportional to the ellipsoid volume) can be proven to be convex on the PD cone $\mathbb{S}_{++}^{N-1}$ [26].

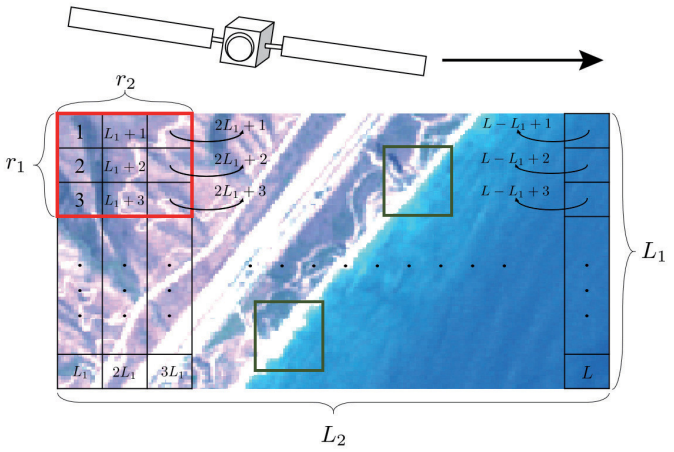

Fig. 1. Pixel indexing of an $L_1 \times L_2$ ROI is according to the acquisition time of the $L = L_1 L_2$ pixels, and exactly corresponds to the column indexing of the M-band hyperspectral data matrix $\boldsymbol{X} \in \mathbb{R}^{M\times L}$. Also, a natural image often contains similar patches as marked by the two green squares, and this self-similarity phenomenon enables our effective decoder design.

B. Measurement Matrix Design

In this section, we design the measurement matrix $\boldsymbol{\Phi}$ for an M-band hyperspectral data $\boldsymbol{X} \in \mathbb{R}^{M\times L}$. For rigorously defining the measurement matrix $\boldsymbol{\Phi}$, we should specify how the pixel vectors $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_L \in \mathbb{R}^M$ are ordered into the columns of $\boldsymbol{X}$, where $L = L_1 L_2$ pixels are with respect to a given $L_1 \times L_2$ region of interest (ROI). It is natural to index the pixels according to their acquisition time, as illustrated in Fig. 1, where L_2 lines (along-track) are sequentially scanned by the pushbroom system onboard the satellite with each line containing L_1 pixels (across-track). According to (1), we design the measurement matrix with respect to the vectorized image $\boldsymbol{x} = \text{vec}(\boldsymbol{X})$ as

$$\begin{aligned}\boldsymbol{\Phi} = & \begin{pmatrix} \boldsymbol{I}_L \otimes \text{bdiag}\left(\boldsymbol{1}_{M_1}^T, \ldots, \boldsymbol{1}_{M_K}^T\right) \\ \dfrac{\boldsymbol{I}_{L_2/r_2} \otimes \boldsymbol{1}_{r_2}^T}{r_2} \otimes \dfrac{\boldsymbol{I}_{L_1/r_1} \otimes \boldsymbol{1}_{r_1}^T}{r_1} \otimes \boldsymbol{I}_M \end{pmatrix} \\ & \in \mathbb{R}^{\left(LK + \frac{ML}{r_1 r_2}\right)\times(ML)}\end{aligned} \tag{6}$$

where $r_1, r_2 \in \mathbb{Z}_{++}$ are integer-valued parameters satisfying $(L_1/r_1), (L_2/r_2) \in \mathbb{Z}_{++}$, and $K, M_1, \ldots, M_K \in \mathbb{Z}_{++}$ are also integer-valued parameters satisfying $M_1 + \cdots + M_K = M$. As will be discussed in Section II-D, this measurement strategy trickily leaves sufficient information in the compressed acquisition $\boldsymbol{y} = \boldsymbol{\Phi}\boldsymbol{x}$ for effective decoding.

To see the physical meaning of the design, as well as that of the integer-valued parameters, we remark that the HCS performed by $\boldsymbol{\Phi}$ actually just involves addition operators (without involving random sampling, projection, or multiplication operators). Also, unlike some joint HCS approaches [25], the proposed design does not require globally collecting complete information of $\boldsymbol{X}$ before compression. Instead, through some basic properties of the Kronecker product, one can easily verify that the upper half of $\boldsymbol{\Phi}$ just performs local addition along the spectral dimension, while the lower half performs local addition along the spatial dimension. To be more specific, the upper half actually partitions the M spectral bands into K groups, followed by adding up the M_i band values in the

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ith group for all $i \in \mathcal{I}_K$; similarly, the lower half essentially partitions the $L_1 \times L_2$ ROI into subregions of size $r_1 \times r_2$ (see Fig. 1), followed by adding up the $r_1 r_2$ pixel values in each subregion.

As discussed, to exploit the recent advances in metamaterials for miniaturized satellite applications, an all-addition HCS encoder is appealing. However, such a simple encoding scheme (6) surely induces a highly challenging design of decoder for recovering $\boldsymbol{x}$ from the measurements $\boldsymbol{y} = \boldsymbol{\Phi x}$, to be developed in Section II-D. Such decoder design is intuitively possible as the compressed acquisition $\boldsymbol{y}$, from the above illustration of physical meaning, still contains the spatial information of $\boldsymbol{x}$ in the upper half (after spectral addition) and spectral information of $\boldsymbol{x}$ in the lower half (after spatial addition). In Section II-D, instead of adopting the TV-regularized decoder popularly adopted in HCS [16], [17], we will show that an arguably stronger regularization based on a convex self-similarity function does make an effective decoder design possible.

C. Basis Matrix Design

The most commonly seen basis matrices for sparsifying signals in CS would be based on the discrete wavelet transform, the Fourier transform, and the discrete cosine transform (DCT). However, all these transforms do not exploit the fundamental nature of HSI data, which is known to lie within a very low-dimensional subspace of $\mathbb{R}^M$, i.e., the column space of the signature matrix $\boldsymbol{A} \in \mathbb{R}^{M \times N}$ [32]. This can be easily seen from the aforementioned model $\boldsymbol{X} = \boldsymbol{AS}$. Thus, naturally, the hyperspectral signals can be sparsified by representing $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_L \in \mathbb{R}^M$ with respect to such low-dimensional column space, denoted as $\mathrm{col}(\boldsymbol{A})$. Let $\boldsymbol{E} \in \mathbb{R}^{M \times N}$ be a semiunitary matrix satisfying $\mathrm{col}(\boldsymbol{E}) = \mathrm{col}(\boldsymbol{A})$. Then, we have $\boldsymbol{X} = \boldsymbol{E}\widetilde{\boldsymbol{S}}$ for some (probably dense) coefficient matrix $\widetilde{\boldsymbol{S}} \triangleq [\widetilde{\boldsymbol{s}}_1, \ldots, \widetilde{\boldsymbol{s}}_L] \in \mathbb{R}^{N \times L}$, thereby leading to the simple and natural design of basis matrix

$$\boldsymbol{\Psi} \triangleq \boldsymbol{I}_L \otimes \boldsymbol{E} \tag{7}$$

which is derived based on the observation that $\boldsymbol{x} = \mathrm{vec}(\boldsymbol{X}) = (\boldsymbol{I}_L \otimes \boldsymbol{E})\widetilde{\boldsymbol{s}} = \boldsymbol{\Psi}\widetilde{\boldsymbol{s}}$ with $\widetilde{\boldsymbol{s}} \triangleq \mathrm{vec}(\widetilde{\boldsymbol{S}})$.

To see why this basis matrix does yield highly sparse signal, note that there exists a semiunitary matrix $\boldsymbol{E}^{\mathrm{C}} \in \mathbb{R}^{M \times (M-N)}$ such that $[\boldsymbol{E},\ \boldsymbol{E}^{\mathrm{C}}] \in \mathbb{R}^{M \times M}$ is unitary. Hence, all the pixel vectors $\boldsymbol{x}_i$ can be represented with respect to $[\boldsymbol{E},\ \boldsymbol{E}^{\mathrm{C}}]$, i.e., $\boldsymbol{x}_i = [\boldsymbol{E},\ \boldsymbol{E}^{\mathrm{C}}]\boldsymbol{s}_i'$ for some coefficients $\boldsymbol{s}_i'$. In fact, it is not difficult to infer the closed-form expression of $\boldsymbol{s}_i' = [\widetilde{\boldsymbol{s}}_i^T,\ \boldsymbol{0}_{M-N}^T]^T$, which is clearly highly sparse as the number of spectral bands M is much larger than the number of materials N in typical HSI. In other words, the sparsity has been automatically enforced in our design (7), and that is why we will not adopt the ℓ_1-norm of BP in regularizing our decoding criterion (to be designed in Section II-D).

We conclude this section with some remarkable properties of the proposed design of $\boldsymbol{\Psi}$. First, we will experimentally illustrate that the natural design of (7), indeed, represents HSI data in a more sparse and precise manner than other benchmark transforms, as indicated by smaller spectral reconstruction errors (measured by spectral angle mapper (SAM) [33]) in Fig. 2; note that SAM is a key index for measuring the quality of HSI, as the key value of HSI is its capability of material identification via spectral shape (or spectral signature) [28], [33]. Also, as reported in [16, Sec. II.C], $\boldsymbol{E}$ is available onboard, meaning that $\boldsymbol{\Psi}$ is also available for further compression onboard, leading to an effective sampling rate of $((KL + NL_h)/ML)$ with $L_h \triangleq L/(r_1 r_2)$. As discussed in Section I-A, some HCS work requires the information of $\boldsymbol{A} \in \mathbb{R}^{M \times N}$, implying the requirement of model-order selection for obtaining N. However, estimating N is known to be a daunting task [34] and may not be feasible for satellite applications, highlighting another great advantage of the proposed $\boldsymbol{\Psi}$ that does not require a precise model-order N; we set $N = 10$ for all our experiments in Section III.

D. Decoding Criterion and Algorithm Design

By (1), (2), and $\boldsymbol{\Theta} = \boldsymbol{\Phi\Psi}$, typical decoder in CS is either to solve $\min_{\widetilde{\boldsymbol{s}}} (1/2)\|\boldsymbol{y} - \boldsymbol{\Theta}\widetilde{\boldsymbol{s}}\|_2^2 + \lambda \mathrm{TV}(\widetilde{\boldsymbol{s}})$ or to solve $\min_{\widetilde{\boldsymbol{s}}} (1/2)\|\boldsymbol{y} - \boldsymbol{\Theta}\widetilde{\boldsymbol{s}}\|_2^2 + \lambda \|\widetilde{\boldsymbol{s}}\|_1$, where total-variation or sparsity regularizers are critical for the ill-posed decoding problem, and $\lambda > 0$ is the regularization parameter. Another good choice of regularizer is based on self-similarity (see Fig. 1), which is strong partly because it has recently been defined as a convex function ϕ and partly because it is scene-adapted [22]. Note that TV is convex but not scene-adapted, as its definition remains unchanged for images with different texture information. Intuitively, a scene-adapted regularizer will yield generally better performance, so self-similarity is considered in our decoder criterion, i.e.,

$$\min_{\widetilde{\boldsymbol{s}}} \ \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{\Theta}\widetilde{\boldsymbol{s}}\|_2^2 + \lambda \phi(\widetilde{\boldsymbol{s}}) \tag{8}$$

and experimental evidence will show the superiority of the self-similarity regularizer ϕ, as defined in (9); note also that we do not adopt the ℓ_1-norm regularizer because the sparsity has been automatically enforced in the design of $\boldsymbol{\Psi}$, as discussed in Section II-C.

Though self-similarity provides strong spatial structural information for various inverse imaging problems, the concept of self-similarity has never been explicitly defined as a mathematical function until very recently. In recent multiresolution analysis literature [22], self-similarity has not only been defined explicitly but has been defined as a convex function. Such convexity is critical as it allows adopting the powerful convex optimization theory for effective decoding, like the majority of CS literature. Though the convex self-similarity in [22] is defined for multiresolution analysis, its philosophy can be adopted to customize the regularizer $\phi(\widetilde{\boldsymbol{s}})$ for our HCS decoder. Specifically, the general form is designed as

$$\phi(\widetilde{\boldsymbol{s}}) \triangleq \frac{1}{2}\sum_{b=1}^{N} \sum_{(i,j) \in \mathcal{G}} \alpha_{i,j} \|\boldsymbol{P}_i \boldsymbol{t}_b - \boldsymbol{P}_j \boldsymbol{t}_b\|_2^2 \tag{9}$$

meaning that the ith and jth patches (collected in the self-similarity graph $\mathcal{G}$) of the image $\widetilde{\boldsymbol{s}} = \mathrm{vec}(\widetilde{\boldsymbol{S}})$ (see Section II-C) are forced to be similar. To be rigorous, the L-pixel image $\boldsymbol{t}_b \in \mathbb{R}^L$ denotes the bth band of $\widetilde{\boldsymbol{S}} \in \mathbb{R}^{N \times L}$, i.e., $\mathrm{vec}(\widetilde{\boldsymbol{S}}^T) = [\boldsymbol{t}_1^T, \ldots, \boldsymbol{t}_N^T]^T$; $\boldsymbol{P}_k \in \mathbb{R}^{q^2 \times L}$ serves as an operator that sifts the

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kth patch of size $q \times q$ from the L-pixel image $\boldsymbol{t}_b$, implying that $\boldsymbol{P}_k$ is highly column-sparse having just q^2 nonzero columns; the self-similarity graph $\mathcal{G}$, collecting pairs of indices of similar patches, can be learned from the compressed acquisition $\boldsymbol{y}$, as will be detailed later; and $\alpha_{i,j} \geq 0$ is the weight of edge (i, j) in the graph $\mathcal{G}$, measuring the similarity between the ith and jth patches, as will also be detailed later.

Simply speaking, the regularizer (9) is to promote solutions $\boldsymbol{t}_b$ whose ith patch $\boldsymbol{P}_i\boldsymbol{t}_b$ and jth patch $\boldsymbol{P}_j\boldsymbol{t}_b$ hold high degree of similarity; higher value of $\alpha_{i,j}$ encourages higher degree of similarity. The questions are how we can know which patches that are similar (i.e., learning the graph $\mathcal{G}$) and how similar they are (i.e., learning the weight $\alpha_{i,j}$). In fact, the required information is retained in the compressed acquisition $\boldsymbol{y}$.

To see it, recall from (6) that the upper half of $\boldsymbol{y}$, i.e., $\overline{\boldsymbol{y}} \triangleq (\boldsymbol{I}_L \otimes \text{bdiag}(\mathbf{1}_{M_1}^T, \ldots, \mathbf{1}_{M_K}^T))\boldsymbol{x} \in \mathbb{R}^{LK}$, can be used to form the reference image defined as $\boldsymbol{y}_{\text{ref}} \triangleq \text{vec}_{K\times L}^{-1}(\overline{\boldsymbol{y}})^T\mathbf{1}_K$, from which the self-similarity pattern $(\mathcal{G}, \{\alpha_{i,j}\})$ can be learned. The physical meaning of the reference image is simply the panchromatic image associated with the HSI $\boldsymbol{x} = \text{vec}(\boldsymbol{X})$, as one can verify that $\boldsymbol{y}_{\text{ref}} = \mathbf{1}_M^T\boldsymbol{X}$ is just to add up all the M spectral bands of $\boldsymbol{X}$, meaning that the reference image is of high quality and suitable for learning the self-similarity pattern. Basically, the similarity measure $\alpha_{i,j}$ should be nonnegative and be a decreasing function of the distance between the ith patch (i.e., $\boldsymbol{P}_i\boldsymbol{y}_{\text{ref}}$) and the jth patch (i.e., $\boldsymbol{P}_j\boldsymbol{y}_{\text{ref}}$) in the reference image. Specifically, the weight defined in [22] is refined as follows for avoiding possible numerical errors:

$$\alpha_{i,j} \triangleq \frac{1}{\|\boldsymbol{P}_i\boldsymbol{y}_{\text{ref}} - \boldsymbol{P}_j\boldsymbol{y}_{\text{ref}}\|_2 + \epsilon}$$

where $\epsilon > 0$ is a small positive constant (set as 10^{-8} in this article) for avoiding ϕ dominating the data-fitting term; $\alpha_{i,j}$'s are then normalized so that the maximum is one. The self-similarity graph is, hence, learned as $\mathcal{G} \triangleq \{(i, j) \mid i \in \mathcal{I}_{L'},\ j \in \arg\max_{j'\neq i} \alpha_{i,j'}\}$, where L' is the number of patches covering $\boldsymbol{y}_{\text{ref}}$; in other words, for each patch i, its most similar (but distinct) patch j is searched to form an edge (i, j) to be added in the self-similarity graph $\mathcal{G}$. As will be seen, this simple graph learning procedure is quite effective in building the self-similarity pattern for regularizing our HCS reconstruction problem. We remark that, for any weighted graph $(\mathcal{G}, \{\alpha_{i,j}\})$ learned above, the self-similarity regularizer ϕ defined in (9) is convex, as the composition of a convex function and an affine function preserves convexity.

Next, we design a computationally efficient decoding algorithm for solving the convex optimization problem (8). Since (8) is of very large-scale, alternating direction method of multipliers (ADMM) is a suitable optimizer for solving it [26]. We begin by explicitly reformulating (8) into the ADMM-form, i.e.,

$$\begin{aligned}\min_{\widetilde{\boldsymbol{s}},\boldsymbol{u}_1,\boldsymbol{u}_2,\boldsymbol{U}_{i,j}} \ & \frac{1}{2}\|\overline{\boldsymbol{y}} - \overline{\boldsymbol{\Phi}}\boldsymbol{u}_1\|_2^2 + \frac{1}{2}\|\underline{\boldsymbol{y}} - \underline{\boldsymbol{\Phi}}\boldsymbol{u}_2\|_2^2 \\ & + \sum_{(i,j)\in\mathcal{G}} \frac{\lambda\alpha_{i,j}}{2}\|(\boldsymbol{P}_i - \boldsymbol{P}_j)\boldsymbol{U}_{i,j}\|_F^2 \\ \text{s.t. } & \boldsymbol{u}_1 = \boldsymbol{\Psi}\widetilde{\boldsymbol{s}},\quad \boldsymbol{u}_2 = \boldsymbol{\Psi}\widetilde{\boldsymbol{s}},\ \boldsymbol{U}_{i,j} = \widetilde{\boldsymbol{S}}^T \end{aligned} \tag{10}$$

Algorithm 1 AAHCSD Algorithm for Solving (10)

1: **Given** $\boldsymbol{y}$, $\lambda > 0$ and $\mu > 0$.
2: Initialize $\boldsymbol{s}^0 := \mathbf{0}_{NL}$, $\boldsymbol{d}_1^0 := \mathbf{0}_{LM}$, $\boldsymbol{d}_2^0 := \mathbf{0}_{LM}$, $\boldsymbol{D}_{i,j}^0 := \mathbf{0}_{L\times N}$ (or warm start). Set $k := 0$.
3: **repeat**
4: Update $\left(\boldsymbol{u}_1^{k+1}, \boldsymbol{u}_2^{k+1}, \left\{\boldsymbol{U}_{i,j}^{k+1}\right\}\right) \in \arg\min_{\boldsymbol{u}_1,\boldsymbol{u}_2,\boldsymbol{U}_{i,j}} \mathcal{L}\big(\boldsymbol{u}_1, \boldsymbol{u}_2, \{\boldsymbol{U}_{i,j}\}, \boldsymbol{s}^k, \boldsymbol{d}_1^k, \boldsymbol{d}_2^k, \{\boldsymbol{D}_{i,j}^k\}\big)$;
5: Update $\boldsymbol{s}^{k+1} \in \arg\min_{\boldsymbol{s}} \mathcal{L}\Big(\boldsymbol{u}_1^{k+1}, \boldsymbol{u}_2^{k+1}, \{\boldsymbol{U}_{i,j}^{k+1}\}, \boldsymbol{s}, \boldsymbol{d}_1^k, \boldsymbol{d}_2^k, \{\boldsymbol{D}_{i,j}^k\}\Big)$;
6: Update $\boldsymbol{d}_1^{k+1} := \boldsymbol{d}_1^k + \boldsymbol{u}_1^{k+1} - \boldsymbol{\Psi}\boldsymbol{s}^{k+1}$, $\boldsymbol{d}_2^{k+1} := \boldsymbol{d}_2^k + \boldsymbol{u}_2^{k+1} - \boldsymbol{\Psi}\boldsymbol{s}^{k+1}$, and $\boldsymbol{D}_{i,j}^{k+1} := \boldsymbol{D}_{i,j}^k + \boldsymbol{U}_{i,j}^{k+1} - (\boldsymbol{S}^{k+1})^T$;
7: $k := k + 1$;
8: **until** the predefined stopping criterion is met.
9: **Output** the reconstructed data $\widehat{\boldsymbol{x}} := \boldsymbol{\Psi}\boldsymbol{s}^k$.

where $\overline{\boldsymbol{y}} \triangleq [\boldsymbol{y}]_{1:LK}$, $\overline{\boldsymbol{\Phi}} \triangleq \boldsymbol{I}_L \otimes \text{bdiag}(\mathbf{1}_{M_1}^T, \ldots, \mathbf{1}_{M_K}^T)$, $\underline{\boldsymbol{y}} \triangleq [\boldsymbol{y}]_{LK+1:LK+(ML/r_1r_2)}$, and $\underline{\boldsymbol{\Phi}} \triangleq ((\boldsymbol{I}_{L_2/r_2} \otimes \mathbf{1}_{r_2}^T)/r_2) \otimes ((\boldsymbol{I}_{L_1/r_1} \otimes \mathbf{1}_{r_1}^T)/r_1) \otimes \boldsymbol{I}_M$. The augmented Lagrangian of (10) is

$$\begin{aligned}&\mathcal{L}(\boldsymbol{u}_1, \boldsymbol{u}_2, \{\boldsymbol{U}_{i,j}\}, \widetilde{\boldsymbol{s}}, \boldsymbol{d}_1, \boldsymbol{d}_2, \{\boldsymbol{D}_{i,j}\}) \\ &= \sum_{(i,j)\in\mathcal{G}}\left(\frac{\lambda\alpha_{i,j}}{2}\|(\boldsymbol{P}_i - \boldsymbol{P}_j)\boldsymbol{U}_{i,j}\|_F^2 + \frac{\mu}{2}\|\widetilde{\boldsymbol{S}}^T - \boldsymbol{U}_{i,j} - \boldsymbol{D}_{i,j}\|_F^2\right) \\ &\quad + \frac{1}{2}\|\overline{\boldsymbol{y}} - \overline{\boldsymbol{\Phi}}\boldsymbol{u}_1\|_2^2 + \frac{\mu}{2}\|\boldsymbol{\Psi}\widetilde{\boldsymbol{s}} - \boldsymbol{u}_1 - \boldsymbol{d}_1\|_2^2 \\ &\quad + \frac{1}{2}\|\underline{\boldsymbol{y}} - \underline{\boldsymbol{\Phi}}\boldsymbol{u}_2\|_2^2 + \frac{\mu}{2}\|\boldsymbol{\Psi}\widetilde{\boldsymbol{s}} - \boldsymbol{u}_2 - \boldsymbol{d}_2\|_2^2\end{aligned}$$

where $\boldsymbol{d}_1$, $\boldsymbol{d}_2$, and $\{\boldsymbol{D}_{i,j}\}$ are scaled dual variables, and $\mu > 0$ is called penalty parameter. The ADMM algorithm for solving (10) is detailed in Algorithm 1, referred to as all-addition HCS decoder (AAHCSD), where $\boldsymbol{s}^k = \text{vec}(\boldsymbol{S}^k)$ denotes the kth iterative update of $\widetilde{\boldsymbol{s}}$. The convergence guarantee of AAHCSD is stated in Theorem 1 and proved in Appendix A.

Theorem 1: Consider any graph $\mathcal{G} \subseteq \{(i, j) \mid i, j \in \mathcal{I}_{L'}, i \neq j\}$ and $\alpha_{i,j} \in (0, 1]\ \forall (i, j) \in \mathcal{G}$, where L' is the number of patches. As long as the penalty parameter $\mu > 0$, the sequence $\{\boldsymbol{s}^k\}$ generated by Algorithm 1 converges to an optimal solution of the self-similarity regularized decoding problem (10). □

To implement Algorithm 1, we still need to derive $\boldsymbol{u}_1^{k+1}$, $\boldsymbol{u}_2^{k+1}$, $\{\boldsymbol{U}_{i,j}^{k+1}\}$, and $\boldsymbol{s}^{k+1}$. We reformulate the decoding problem in the form of (10) not just for matching the ADMM-form but also for allowing computationally efficient closed-form solutions. Specifically, the three variables in the ADMM-update of $(\boldsymbol{u}_1, \boldsymbol{u}_2, \{\boldsymbol{U}_{i,j}\})$ (i.e., Line 4 of Algorithm 1) can be decoupled, allowing us to separately handling them.

We first handle $\boldsymbol{u}_1^{k+1}$, which can be simplified as

$$\begin{aligned}\boldsymbol{u}_1^{k+1} &= \arg\min_{\boldsymbol{u}_1}\ \|\overline{\boldsymbol{y}} - (\boldsymbol{I}_L \otimes \boldsymbol{D})\boldsymbol{u}_1\|_2^2 + \mu\|\boldsymbol{v}_1 - \boldsymbol{u}_1\|_2^2 \\ &= [\boldsymbol{I}_L \otimes (\boldsymbol{D}^T\boldsymbol{D}) + \mu\boldsymbol{I}_{LM}]^{-1}[(\boldsymbol{I}_L \otimes \boldsymbol{D}^T)\overline{\boldsymbol{y}} + \mu\boldsymbol{v}_1] \quad (11) \\ &= [\boldsymbol{I}_L \otimes (\boldsymbol{D}^T\boldsymbol{D} + \mu\boldsymbol{I}_M)^{-1}][(\boldsymbol{I}_L \otimes \boldsymbol{D}^T)\overline{\boldsymbol{y}} + \mu\boldsymbol{v}_1] \quad (12)\end{aligned}$$

where $\boldsymbol{D} \triangleq \text{bdiag}(\mathbf{1}_{M_1}^T, \ldots, \mathbf{1}_{M_K}^T) \in \mathbb{R}^{K\times M}$ and $\boldsymbol{v}_1 \triangleq \boldsymbol{\Psi}\boldsymbol{s}^k - \boldsymbol{d}_1^k$. Equation (12) provides a computationally efficient solution as it only involves a matrix inversion of much lower dimensionality; note that if the dimension depends on the number of pixels

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L (e.g., (11)), the matrix inversion is not computationally feasible in practice.

Second, we handle $\boldsymbol{u}_2^{k+1}$, which can be written as

$$\begin{aligned}\boldsymbol{u}_2^{k+1} &= \arg\min_{\boldsymbol{u}_2} \ \|\underline{\boldsymbol{y}} - (\boldsymbol{B}^T \otimes \boldsymbol{I}_M)\boldsymbol{u}_2\|_2^2 + \mu\|\boldsymbol{v}_2 - \boldsymbol{u}_2\|_2^2 \\ &= [(\boldsymbol{B} \otimes \boldsymbol{I}_M)(\boldsymbol{B}^T \otimes \boldsymbol{I}_M) + \mu \boldsymbol{I}_{LM}]^{-1}[(\boldsymbol{B} \otimes \boldsymbol{I}_M)\underline{\boldsymbol{y}} + \mu \boldsymbol{v}_2] \\ &= [(\boldsymbol{B}\boldsymbol{B}^T + \mu \boldsymbol{I}_L)^{-1} \otimes \boldsymbol{I}_M][(\boldsymbol{B} \otimes \boldsymbol{I}_M)\underline{\boldsymbol{y}} + \mu \boldsymbol{v}_2]\end{aligned}$$

where $\boldsymbol{B} \triangleq (1/r)(\boldsymbol{I}_{L_2/r_2} \otimes \boldsymbol{1}_{r_2}) \otimes (\boldsymbol{I}_{L_1/r_1} \otimes \boldsymbol{1}_{r_1}) \in \mathbb{R}^{L \times L_h}$, $\boldsymbol{v}_2 \triangleq \boldsymbol{\Psi}\boldsymbol{s}^k - \boldsymbol{d}_2^k$, and $r \triangleq r_1 r_2$. However, this is still not computationally affordable due to the $L \times L$ matrix inversion. To reduce the complexity, by observing the structure of $\boldsymbol{B}$, it is not difficult to find that there exists a permutation matrix $\boldsymbol{\Pi}$ such that $\boldsymbol{B} = (1/r)\boldsymbol{\Pi}(\boldsymbol{I}_{L_h} \otimes \boldsymbol{1}_r)$. Thus, we have

$$(\boldsymbol{B}\boldsymbol{B}^T + \mu \boldsymbol{I}_L)^{-1} = \boldsymbol{\Pi}\big[\boldsymbol{I}_{L_h} \otimes \big(\boldsymbol{1}_r\boldsymbol{1}_r^T/r^2 + \mu \boldsymbol{I}_r\big)^{-1}\big]\boldsymbol{\Pi}^T \quad (13)$$

with which $\boldsymbol{u}_2^{k+1}$ can be efficiently computed as only a low-dimensional $r \times r$ matrix inversion is involved.

Third, $\boldsymbol{U}_{i,j}^{k+1} = \arg\min_{\boldsymbol{U}_{i,j}} (\lambda\alpha_{i,j}/2)\|(\boldsymbol{P}_i - \boldsymbol{P}_j)\boldsymbol{U}_{i,j}\|_F^2 + (\mu/2)\|(\boldsymbol{S}^k)^T - \boldsymbol{U}_{i,j} - \boldsymbol{D}_{i,j}^k\|_F^2$ is handled by observing that it can be decoupled columnwise, i.e.,

$$\min_{\boldsymbol{u}_{i,j}^{(c)}} \ \frac{\lambda\alpha_{i,j}}{2}\big\|(\boldsymbol{P}_i - \boldsymbol{P}_j)\boldsymbol{u}_{i,j}^{(c)}\big\|_2^2 + \frac{\mu}{2}\big\|\boldsymbol{u}_{i,j}^{(c)} - \boldsymbol{\delta}_{i,j}^{(c)}\big\|_2^2 \quad (14)$$

where $\boldsymbol{\delta}_{i,j}^{(c)}$ denotes the cth column of $(\boldsymbol{S}^k)^T - \boldsymbol{D}_{i,j}^k$, and $\boldsymbol{u}_{i,j}^{(c)}$ is the cth column of $\boldsymbol{U}_{i,j}^{k+1}$. Directly solving (14) requires computing the inverse of a large $L \times L$ matrix, not computationally affordable, but one can trickily exploit the column sparsity of $\boldsymbol{P}_i - \boldsymbol{P}_j \in \mathbb{R}^{q^2 \times L}$ to reduce the complexity; note that $\boldsymbol{P}_i - \boldsymbol{P}_j$ has maximally $2q^2 \ll L$ nonzero columns ($q = 15$ in this work), allowing us to further decouple variables of (14) into two parts, i.e.,

$$\begin{aligned}\min_{\left[\boldsymbol{u}_{i,j}^{(c)}\right]_{\mathcal{I}}, \left[\boldsymbol{u}_{i,j}^{(c)}\right]_{\bar{\mathcal{I}}}} \ & \frac{\lambda\alpha_{i,j}}{2}\big\|[\boldsymbol{P}_i - \boldsymbol{P}_j]_{\mathcal{I}}\big[\boldsymbol{u}_{i,j}^{(c)}\big]_{\mathcal{I}}\big\|_2^2 \\ & + \frac{\mu}{2}\big\|\big[\boldsymbol{u}_{i,j}^{(c)}\big]_{\mathcal{I}} - \big[\boldsymbol{\delta}_{i,j}^{(c)}\big]_{\mathcal{I}}\big\|_2^2 + \frac{\mu}{2}\big\|\big[\boldsymbol{u}_{i,j}^{(c)}\big]_{\bar{\mathcal{I}}} - \big[\boldsymbol{\delta}_{i,j}^{(c)}\big]_{\bar{\mathcal{I}}}\big\|_2^2 \quad (15)\end{aligned}$$

where $\mathcal{I} \subseteq \mathcal{I}_L$ is the index set collecting indices of nonzero columns of the matrix $(\boldsymbol{P}_i - \boldsymbol{P}_j)$, while $\bar{\mathcal{I}} \triangleq \mathcal{I}_L \setminus \mathcal{I}$ is the complement of $\mathcal{I}$; the notation $[\boldsymbol{v}]_{\mathcal{S}}$ denotes the subvector of $\boldsymbol{v}$ formed by collecting entries with indices specified by the index set $\mathcal{S}$; and $[\boldsymbol{P}_i - \boldsymbol{P}_j]_{\mathcal{I}} \in \mathbb{R}^{q^2 \times |\mathcal{I}|}$ is then a trimmed version of $(\boldsymbol{P}_i - \boldsymbol{P}_j)$ by removing those zero columns. Note that q is set as 15 in this work, while L is usually much larger than ten thousands, implying that the leading term in (15) is of much lower dimensionality, enabling the following computationally efficient closed-form solutions:

$$\begin{aligned}\big[\boldsymbol{u}_{i,j}^{(c)}\big]_{\mathcal{I}}^{\star} &= \left[[\boldsymbol{P}_i - \boldsymbol{P}_j]_{\mathcal{I}}^T[\boldsymbol{P}_i - \boldsymbol{P}_j]_{\mathcal{I}} + \frac{\mu \boldsymbol{I}_{|\mathcal{I}|}}{\lambda\alpha_{i,j}}\right]^{-1}\left(\frac{\mu\big[\boldsymbol{\delta}_{i,j}^{(c)}\big]_{\mathcal{I}}}{\lambda\alpha_{i,j}}\right) \\ \big[\boldsymbol{u}_{i,j}^{(c)}\big]_{\bar{\mathcal{I}}}^{\star} &= \big[\boldsymbol{\delta}_{i,j}^{(c)}\big]_{\bar{\mathcal{I}}}\end{aligned}$$

where the dimension of the involved matrix inversion is just $|\mathcal{I}| \leq 2q^2 \ll L$.

Finally, we need to handle $\boldsymbol{s}^{k+1}$, which is the $(k+1)$th iterative update of $\widetilde{\boldsymbol{s}} = \mathrm{vec}(\widetilde{\boldsymbol{S}})$. By the fact of $\|\boldsymbol{M}\|_F = \|\boldsymbol{M}^T\|_F$ for any matrix $\boldsymbol{M}$ and the definition of augmented Lagrangian, the subproblem of $\boldsymbol{s}^{k+1}$ (see Line 5 of Algorithm 1) can be explicitly written as

$$\begin{aligned}&\arg\min_{\boldsymbol{s}} \ \big\|\boldsymbol{\Psi}\boldsymbol{s} - \boldsymbol{t}_1'\big\|_2^2 + \big\|\boldsymbol{\Psi}\boldsymbol{s} - \boldsymbol{t}_2'\big\|_2^2 + \sum_{(i,j)\in\mathcal{G}} \|\boldsymbol{s} - \boldsymbol{t}_{i,j}\|_2^2 \\ &\quad = (2\boldsymbol{\Psi}^T\boldsymbol{\Psi} + |\mathcal{G}|\boldsymbol{I}_{NL})^{-1}\left(\boldsymbol{\Psi}^T\boldsymbol{t}_1' + \boldsymbol{\Psi}^T\boldsymbol{t}_2' + \sum_{(i,j)\in\mathcal{G}} \boldsymbol{t}_{i,j}\right)\end{aligned}$$

where $\boldsymbol{t}_1' \triangleq \boldsymbol{u}_1^{k+1} + \boldsymbol{d}_1^k$, $\boldsymbol{t}_2' \triangleq \boldsymbol{u}_2^{k+1} + \boldsymbol{d}_2^k$, and $\boldsymbol{t}_{i,j} \triangleq \mathrm{vec}[(\boldsymbol{U}_{i,j}^{k+1} + \boldsymbol{D}_{i,j}^k)^T]$. The huge (NL)-dimensional matrix inversion can be efficiently computed by exploiting the structure of $\boldsymbol{\Psi}$, leading to an equivalent representation of $(2\boldsymbol{\Psi}^T\boldsymbol{\Psi} + |\mathcal{G}|\boldsymbol{I}_{NL})^{-1} = \boldsymbol{I}_L \otimes (2\boldsymbol{E}^T\boldsymbol{E} + |\mathcal{G}|\boldsymbol{I}_N)^{-1}$, which involves just an N-dimensional matrix inversion. Complexity analysis of Algorithm 1 is summarized in Appendix B, and the design of the decoder has been completed.

E. Analysis and Discussion

Our analysis will be mainly based on the JE framework, recently developed in the mathematics and machine learning literature [10], [23], where the fundamental role of JE in analyzing low-purity and ill-conditioned HSI has been revealed. In view of this, the JE-based analysis would be more practical than the commonly seen RIP-based one. Both results are summarized here.

Theorem 2: If the $(\boldsymbol{\Phi}, \boldsymbol{x})$ pair satisfies RIP with k-sparse $\boldsymbol{x}$, having $\delta_{2k} < \sqrt{2} - 1 \in (0, 1)$, then the convex BP solution yields exact $\widehat{\boldsymbol{x}}$. □

Theorem 3: If the $(\boldsymbol{\Phi}, \boldsymbol{x})$ pair satisfies AIP with γ-pure $\boldsymbol{x}$, having $\gamma > 1/\sqrt{N-1} \in (0, 1]$, then the convex JE solution yields exact $\widehat{\boldsymbol{x}}$. □

Theorem 2 follows from Lemma 2 and the remark below it, while Theorem 3 is proven in Appendix D. Note that, in Definition 3, γ is defined with respect to observable $\boldsymbol{x}$. However, in HCS, $\boldsymbol{x}$ is not observable. Thus, by saying that $\boldsymbol{x}$ is γ-pure, it actually means that the observable quantity $\boldsymbol{\Phi}\boldsymbol{x}$ contains a hyperspectral data set (i.e., $\underline{\boldsymbol{\Phi}\boldsymbol{x}}$) with purity γ.[1]

Theorem 3 is more interesting than Theorem 2, as discussed in the following. Both sufficient exact recovery conditions are not verifiable in practice, so our discussion will be focused on whether the conditions well capture the nature of HSI. For RIP, the required high sparsity seems less practical as typical HSI data represents dense reflectance or radiation values. Note that, without sparsity, the RIP $(1 - \delta_k)\|\boldsymbol{x}\|_2^2 \leq \|\boldsymbol{\Phi}\boldsymbol{x}\|_2^2 \leq (1 + \delta_k)\|\boldsymbol{x}\|_2^2$ is, in general, not true (for all $\delta_k \in (0, 1)$), as can be easily seen from the toy counterexample of dense $\boldsymbol{x} = \boldsymbol{1}_8$ (e.g., $\boldsymbol{\Phi}$ with $K = 1$, $M = M_1 = 2$, $L_1 = L_2 = 2$, and $r_1 = r_2 = 2$, for which $\|\boldsymbol{\Phi}\boldsymbol{x}\|_2^2 = 48$). By contrast, AIP is more practical as indicated by Lemma 3, whose proof is given in Appendix C; by nature of the signature matrix $\boldsymbol{A}$, each of

[1] To provide more details, such observable hyperspectral data set comes from the lower half of $\boldsymbol{\Phi}$ and can be explicitly written as $\underline{\boldsymbol{\Phi}\boldsymbol{x}} = (\boldsymbol{B}^T \otimes \boldsymbol{I}_M)\boldsymbol{x} = \mathrm{vec}(\boldsymbol{X}\boldsymbol{B})$. Note that $\overline{\boldsymbol{X}} \triangleq \boldsymbol{X}\boldsymbol{B} \in \mathbb{R}^{M \times L_h}$ can be viewed as an observable M-band HSI, whose effective abundances $\overline{\boldsymbol{S}}$ (used in (3)) can be derived as $\overline{\boldsymbol{X}} = \boldsymbol{A}\boldsymbol{S}\boldsymbol{B} = \boldsymbol{A}\overline{\boldsymbol{S}}$ with $\overline{\boldsymbol{S}} \triangleq \boldsymbol{S}\boldsymbol{B}$. Similarly, in Theorem 3, the JE also refers to as the maximum-volume ellipsoid inscribed in the observable data convex hull $\mathrm{conv}\{\overline{\boldsymbol{X}}\}$.

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its entry representing reflectance belongs to the interval [0, 1], with which we have the following lemma.

Lemma 3: Assume that entries of $\boldsymbol{A} \in \mathbb{R}^{M\times N}$ are independently drawn from the uniform distribution $\mathcal{U}[0,1]$. If $K \geq N-1$, then the proposed $\boldsymbol{\Phi}$ satisfies AIP with probability one (w.p.1). □

On the other hand, another fact motivating us to investigate Theorem 3 is that, for the HSI data remotely acquired on miniaturized satellite, the data purity γ is usually very low as each remotely sensed pixel usually corresponds to a large spatial area.

To be more specific, recalling the range of $\gamma \in [1/\sqrt{N}, 1]$, the assumption of highest data purity $\gamma = 1$ is necessary in many signal processing techniques, such as the *PPA* in blind source separation [30], the *separability assumption* in machine learning [35], and the *well-grounded point (WGP) assumption* in magnetic resonance imaging [36], all theoretically equivalent to $\gamma = 1$. Nevertheless, remotely acquired HSI data usually has $\gamma < 1$ [8], [24]. This fact is completely ignored in conventional RIP analysis, i.e., Theorem 2. In Theorem 3, however, just a very low purity of $\gamma > 1/\sqrt{N-1}$ is required, having better practical meaning. Another important practical concern is the ill-conditioned signature matrix of typical HSI, caused by similar hyperspectral signatures among different materials [10], as commonly observed from benchmark HSI (e.g., the data set provided by NASA [37]). Again, this is not considered in conventional RIP analysis, while JE adopted in Theorem 3 has been reported to be the key for analyzing such ill-conditioned HSI [10].

III. Experimental Results

In this section, we show that the proposed all-addition HCS method yields state-of-the-art performance even without the conventional RP operators. We begin by illustrating the experiment design, followed by comprehensive experiments conducted on real HSI data sets.

A. Experiment Setting

The proposed AAHCSD method will be compared with several benchmark HCS methods, including HYCA [16], SpeCA [12], LSSHUCS [17], and JTenRe3DTV [25]. Though optimizing the setting from an infinite parameter space is infeasible, we have tried our best to refine the parameters for each algorithm in the most important case of 1% sampling rate. As the refined parameters also perform very well for other sampling rates, they are used in the subsequent experimental investigation, as summarized in Table I. We acknowledge that further refining the parameters would be possible, but we believe that it could just lead to marginal improvements not affecting the conclusions. Also, our research philosophy is that software engineer should find a *unifying* parameter setting for different scenarios, not only to be more practically applicable but also to be user friendly (see, e.g., our our previous work [33] wherein four parameters are unified for all scenarios). In view of this, we unify the parameter setting of AAHCSD as $(\lambda, \mu) = (1, 1)$ though such setting may not be optimal for all cases. We should also clarify that only SpeCA and AAHCSD work blindly, while HYCA and LSSHUCS assume the availability of pure endmember signatures; though JTenRe3DTV seems to work blindly, it adopts a holistic compressive manner requiring the entire HSI cube (see Section I-A), not suitable for miniaturized satellite application.

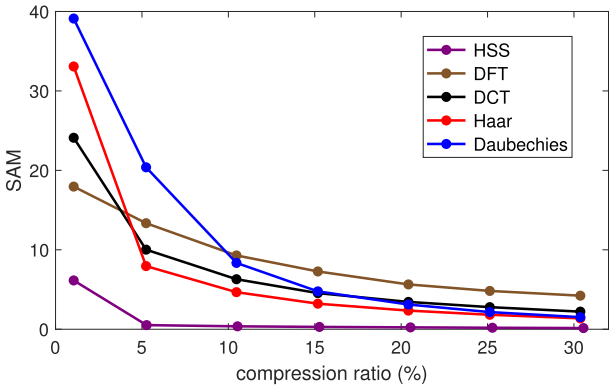

Fig. 2. Comparison between different basis matrices for sparsifying the signals, where SAM is measured in degree.

TABLE I
Implementation Details and Parameter Settings for Different HCS Algorithms

Algorithms	Implementation Details and Parameter Settings
HYCA	window size: 5; penalty parameter: 0.005 regularization parameter: 0.01
SpeCA	size of the first-part RP measurement m_a: AA number of the second-part RP measurement v: AA size of the second-part RP measurement m_b: 1 *(AA: automatically adapted for a target sampling rate)*
LSSHUCS	model order: 20; regularization parameter: 1
JTenRe3DTV	regularization parameter: 0.03

To have a fair comparison, besides qualitative comparison, we also quantitatively compare these methods using three popular quality metrics for measuring the similarity between the reconstructed HSI $\widehat{\boldsymbol{x}}$ and its ground-truth counterpart $\boldsymbol{x}$. These metrics include the peak signal-to-noise ratio (PSNR), SAM, and the root-mean-squared error (RMSE). PSNR and RMSE are well-known metrics, while SAM is particularly designed for HSI. Due to space limitation, we just mention the definition of SAM [33], i.e.,

$$\text{SAM} = \frac{1}{L}\sum_{n=1}^{L} \arccos\left(\frac{\widehat{\boldsymbol{x}}_n^T \boldsymbol{x}_n}{\|\widehat{\boldsymbol{x}}_n\|_2 \cdot \|\boldsymbol{x}_n\|_2}\right)$$

where $\widehat{\boldsymbol{x}}_n$ is the nth reconstructed pixel. Note that SAM is measured in degree, indicating the spectral property preservation in the reconstructed HSI. As the value of HSI is mainly defined by its informative pixel spectra, SAM is regarded as a key performance index in the hyperspectral analysis. The smaller the value of SAM, the better the spectral property preservation of $\widehat{\boldsymbol{x}}$. As for the spatial quality of the reconstructed HSI $\widehat{\boldsymbol{x}}$, it is measured by PSNR (in dB). The larger the value of PSNR, the higher the spatial quality. Finally, we adopt RMSE to measure the global reconstruction quality of $\widehat{\boldsymbol{x}}$. The smaller the value of RMSE, the higher the quality of the reconstructed image.

Since a miniaturized satellite is our target application, the inevitable noise corruption during the transmission from the satellite to the ground station should be considered in

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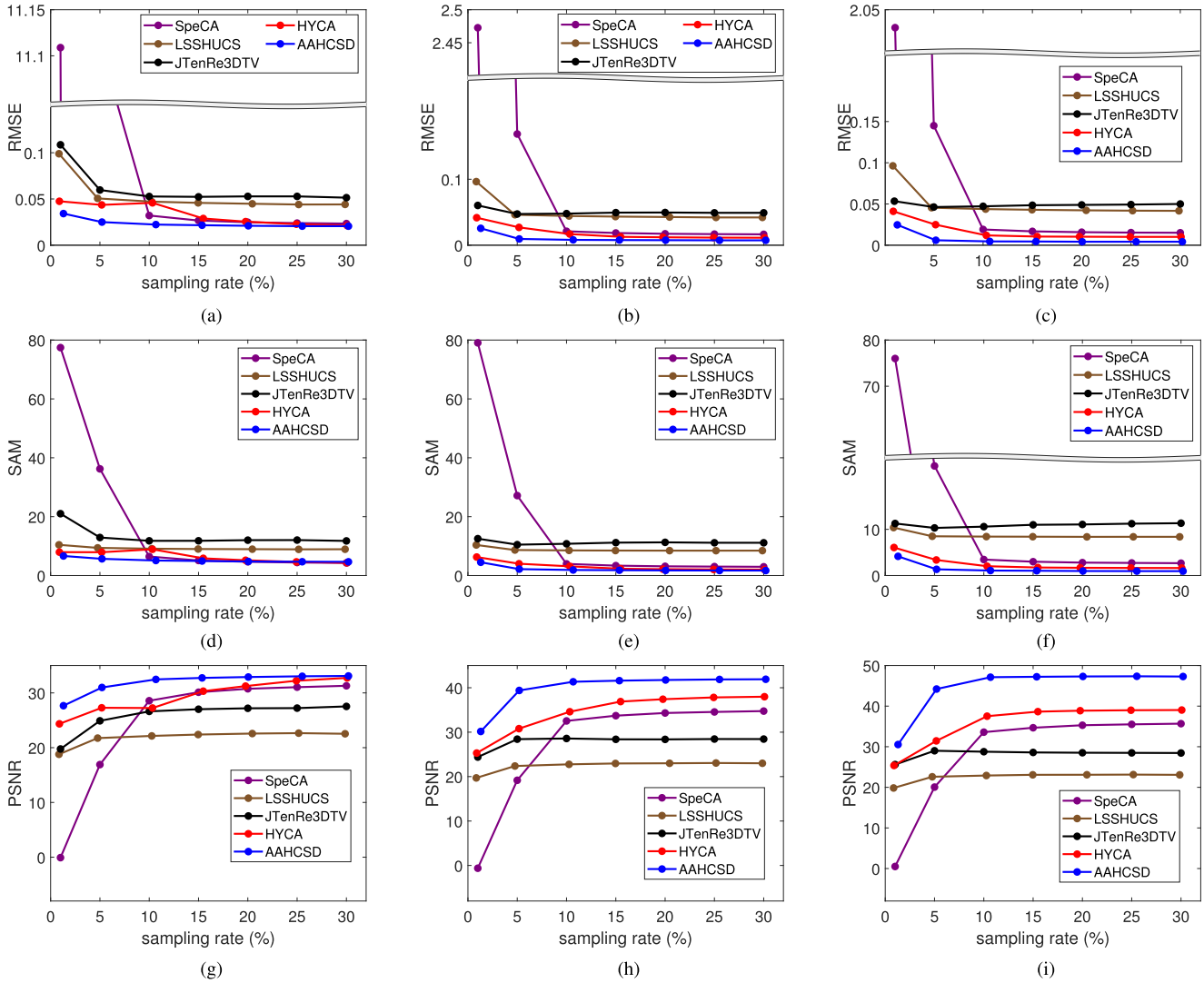

Fig. 3. Performance comparison in terms of RMSE, SAM and PSNR, for different sampling rates and SNR values, using the Chikusei data set. (a) SNR = 25 db. (b) SNR = 35 db. (c) SNR = 45 db. (d) SNR = 25 db. (e) SNR = 35 db. (f) SNR = 45 db. (g) SNR = 25 db. (h) SNR = 35 db. (i) SNR = 45 db.

our experiments. Specifically, the ground-truth HSI $\boldsymbol{x}$ is from two widely studied real hyperspectral data sets that are encoded by various HCS methods to obtain the compressed acquisition $\boldsymbol{y}$. Then, due to the inevitable noise corruption during the transmission, $\boldsymbol{y}$ is further corrupted by the Gaussian noise before it is fed into the decoders of various HCS methods for reconstructing $\widehat{\boldsymbol{x}}$. Specifically, the Gaussian noise is i.i.d. zero-mean with variance σ^2 for various signal-to-noise ratios (SNRs), defined as the ratio between average power of compressed/transmitted data and σ^2. The two benchmark real hyperspectral data sets are introduced as follows.

1) The first data set [38] was acquired by the hyperspectral digital imagery collection experiment (HYDICE) sensor over the Washington DC Mall, USA. A total of $M = 191$ spectral bands, ranging from 400 to 2500 nm, are used in our experimental study. The studied HSI contains $L = 180 \times 180$ pixels in an ROI (see the left column of Fig. 6) commonly investigated in hyperspectral remotes sensing literature [39].
2) The second data set was acquired by Headwall's Hyperspec Visible and Near-Infrared, series C (VNIR-C) imaging sensor over Chikusei, Ibaraki, Japan [40]. A total of $M = 128$ spectral bands, ranging from 363 to 1018 nm, are used in our experiments. A commonly studied $L = 180 \times 180$ ROI is used in this experiment [39], as displayed in the left column of Fig. 5.

For both data sets, AAHCSD performs well under the unifying parameter setting $(\lambda, \mu) = (1, 1)$, as will be seen next. Since an HCS method may not be able to precisely achieve an arbitrarily prespecified sampling rate, we have tried our best to tune the related parameters of the HCS methods under comparison to nearly achieve the prespecified sampling rate.

All the experiments are conducted using Mathworks MATLAB R2019a, and the computer facility is equipped with Core-i7-9700K CPU with 3.60-GHz speed and 32-GB random access memory. Computational time T (measured in seconds) is then adopted as an index of computational efficiency (see Table II).

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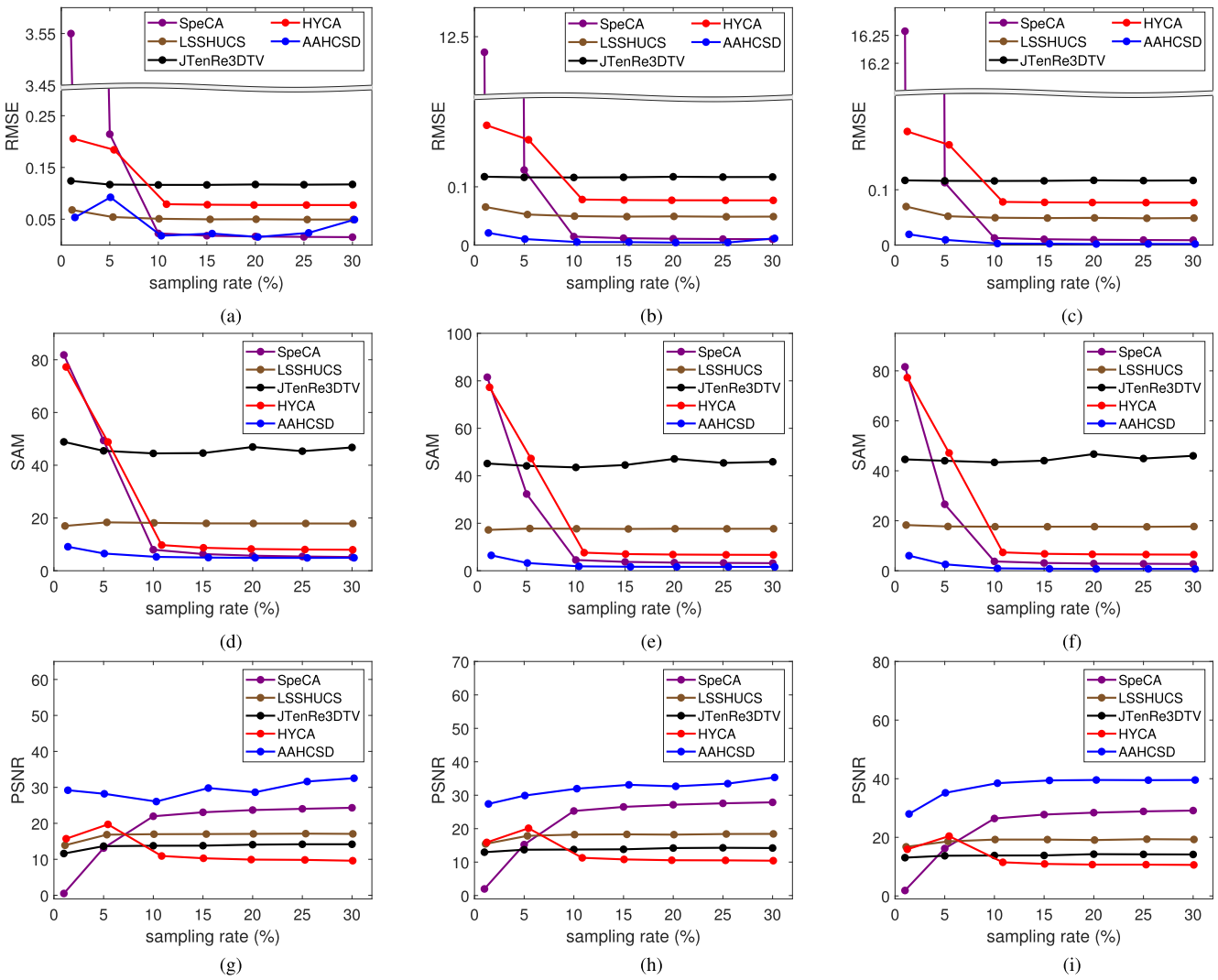

Fig. 4. Performance comparison in terms of RMSE, SAM and PSNR, for different sampling rates and SNR values, using the Washington DC data set. (a) SNR = 25 db. (b) SNR = 35 db. (c) SNR = 45 db. (d) SNR = 25 db. (e) SNR = 35 db. (f) SNR = 45 db. (g) SNR = 25 db. (h) SNR = 35 db. (i) SNR = 45 db.

TABLE II
COMPUTATIONAL TIME T (IN SECONDS) OF DECODERS FOR VARIOUS HCS METHODS

Dataset	Algorithm	Computational Time T For Different Sampling Rates			
		$\approx 1\%$	$\approx 10\%$	$\approx 20\%$	$\approx 30\%$
Washington DC Mall USA	HYCA	14.25	14.58	14.61	14.80
	SpeCA	**0.02937**	**0.2102**	**0.7108**	**1.270**
	LSSHUCS	15.88	20.07	23.16	26.30
	JTenRe3DTV	851.4	944.6	957.1	1098
	AAHCSD	90.16	91.43	88.64	88.77
Chikusei Ibaraki Japan	HYCA	14.08	13.89	14.02	14.32
	SpeCA	**0.01863**	**0.06787**	**0.2724**	**0.5518**
	LSSHUCS	6.861	36.59	36.99	36.99
	JTenRe3DTV	658.4	658.5	660.7	675.6
	AAHCSD	79.06	79.11	79.22	79.26

B. Comparison Between Different Basis Matrices

In this section, we show that the proposed hyperspectral subspace (HSS) basis, indeed, preserves more information of a given HSI data than other commonly adopted basis matrices, including discrete Fourier transform (DFT), DCT, Haar wavelet, and Daubechies wavelet [41]. Specifically, the Washington DC data $\boldsymbol{x}$ introduced in Section III-A were transformed into another domain using the abovementioned transforms. Then, coefficients in the transformed domain are filtered by dropping a prespecified percentage of coefficients with the smallest absolute values, followed by inversely transforming back to the original domain to reconstruct the HSI $\widehat{\boldsymbol{x}}$. Then, the spectral similarity between $\widehat{\boldsymbol{x}}$ and $\boldsymbol{x}$ is measured by SAM.

To evaluate the effectiveness of these transforms, we tested a wide range of compression ratios, i.e., the percentage of dropped coefficients in the transformed domain, and the results are summarized in Fig. 2. As one can see, the suggested HSS basis, indeed, yields the best reconstruction of HSI as indicated by the smallest SAM value at any given compression ratio. Among the other transforms, the probably most well-known Haar wavelet has outstanding performance when the compression ratio is higher than 5%, while DFT has relatively good performance at 1% compression ratio. DCT and Daubechies wavelets also have reasonable performance for compression ratios greater than 15% but still cannot outperform HSS. Therefore, HSS does represent the HSI in a more sparse and precise manner than other widely used benchmark transforms.

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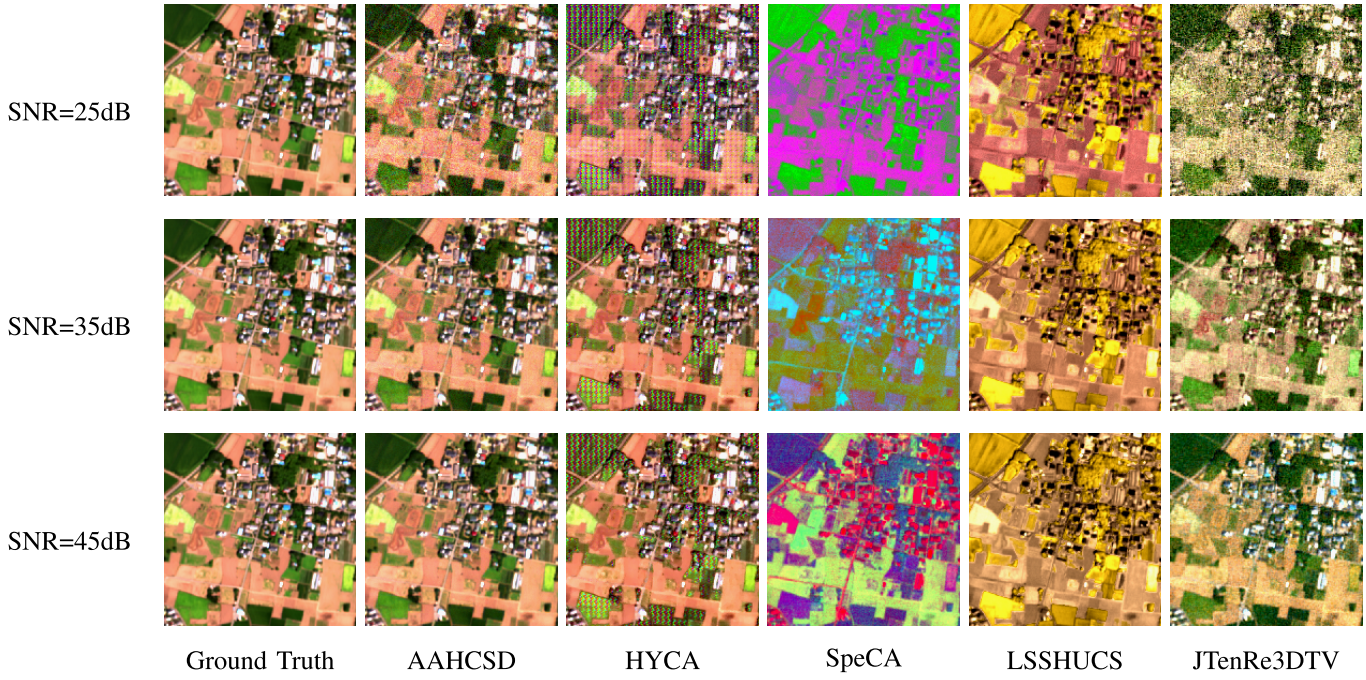

Fig. 5. Qualitative evaluation of various HCS methods at around 5% sampling rate using the color compositions from bands 56 (R), 34 (G) and 19 (B) in the reconstructed Chikusei HSI data $\widehat{\boldsymbol{x}}$.

C. Quantitative Comparison

In this section, we show comprehensive quantitative comparisons between the proposed AAHCSD and other benchmark HCS methods. For each scenario, defined by data set, sampling rate, and SNR, we conduct the experiments with 30 independent Monte Carlo runs and collectively present, in Figs. 3 and 4, the average performances of all the HCS methods in terms of RMSE, SAM, and PSNR. Since our target application is the miniaturized satellite where a huge volume of high-dimensional HSI data is acquired, performance comparison at a very low sampling rate (e.g., $<$10%) is more interesting and needed for efficient data transmission.

AAHCSD always performs best or nearly the best for all scenarios, even if only the addition operators are used in the encoding stage, showing a great advantage when comparing to other conventional RP-based HCS methods. For the Chikusei data set, SpeCA and HYCA have an outstanding performance at higher sampling rates, but their performances are relatively weak for lower sampling rates (see Fig. 3). JTenRe3DTV has relatively weak spectral reconstruction performance for the Washington DC data set probably because its decoding criterion is nonconvex; this observation motivated us to design a convex decoder for the proposed AAHCSD. As can be seen from Fig. 4, for the Washington DC data set, SpeCA and LSSHUCS have outstanding performances at sampling rates higher than 10%, in terms of RMSE and PSNR. Nevertheless, the proposed AAHCSD significantly outperforms other HCS methods especially in terms of PSNR.

From Figs. 3 and 4, one can see that, for both data sets, when noise corruption becomes serious, the performances of all the HCS methods under test become weaker in terms of all the indices. Also, for both data sets, reconstruction of HSI $\widehat{\boldsymbol{x}}$ becomes more challenging at a lower sampling rate, as indicated by higher values of RMSE/SAM and smaller values of PSNR. This natural trend seems violated by HYCA, as can be seen from the PSNR values for the Washington DC data set, probably because a larger sampling rate induces an ill-conditioned $\boldsymbol{K}$ in the decoding criterion of HYCA (see Section I-A). Remarkably, for both data sets, the proposed AAHCSD performs best for all the scenarios at the extremely low sampling rate of 1%, having the significant meaning given the huge volume of typical HSI data.

We conclude this section with the following remark. The very high spatial reconstruction quality of AAHCSD, indicated by the best PSNR values, shows that the self-similarity regularizer is, indeed, a strong spatial prior, which also yields great success in solving other imaging inverse problems (see the block-matching and 3-D filtering (BM3D) technique [42]). The strength of the self-similarity regularizer is further enlarged by defining it as a convex function (see Section II-D) as achieved in recent remote sensing literature [22]. These advances enable our all-addition HCS technique.

D. Qualitative Comparison

Finally, to better understand the spatial reconstruction performance of these HCS methods, we perform qualitative analysis in this section (see Figs. 5 and 6). Since some methods yield uninterpretable results at the 1% sampling rate, we demonstrate the qualitative results for the 5% sampling rate; note that results for sampling rate greater than 10% are less interesting for our target application as it may not be effective to tackle the huge volume of typical HSI data.

For each HCS method and a given SNR, the color composition from bands 56, 34, and 19 in the reconstructed Chikusei data $\widehat{\boldsymbol{x}}$ is used to synthesize the RGB images for visualization, as demonstrated in Fig. 5. Similarly, the color

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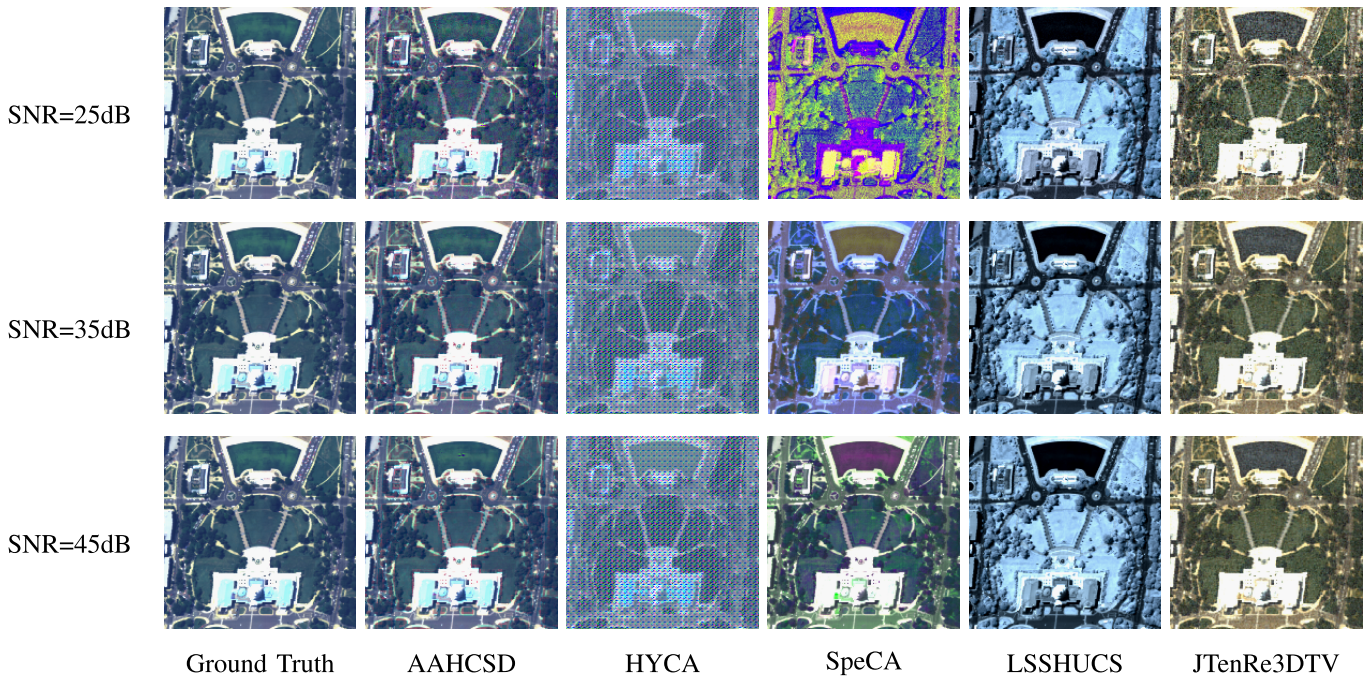

Fig. 6. Qualitative evaluation of various HCS methods at around 5% sampling rate using the color compositions from bands 51 (R), 41 (G) and 22 (B) in the reconstructed Washington DC HSI data $\widehat{\boldsymbol{x}}$.

composition from bands 51, 41, and 22 in the reconstructed Washington DC data $\widehat{\boldsymbol{x}}$ is used to synthesize the RGB images, as demonstrated in Fig. 6. As one can see from these figures, only the proposed AAHCSD has correctly decoded the spatial details for both data sets, as can be seen from the high resemblance between the ground-truth images and the reconstructed HSI. All the other HCS methods either suffer from serious color distortion or spatial detail misreconstruction, for both data sets. With this challenging scenario of 5% sampling rate, the high spatial quality of the reconstructed $\widehat{\boldsymbol{x}}$ again shows the effectiveness of the proposed convex self-similarity decoder adopted in the AAHCSD.

Nevertheless, we have to mention that the good performance of AAHCSD is at the cost of longer computational time, as displayed in Table II. Though we have tried our best to develop computationally efficient closed-from solutions for all the encountered huge dimensional matrix inversions (see Section II-D), the proximal operator induced by the self-similarity function (9) in ADMM still requires further investigation. Once such a proximal operator can be handled very fast, we believe that our convex self-similarity regularization has the potential to replace the seminal TV regularization in several imaging inverse problems [22]. Fortunately, in our target application, the decoding stage is conducted on the ground station, and real-time decoding does not seem to be necessary.

IV. Conclusion and Future Work

Motivated by recent advances in nanoscale multifunctional metasurface, which is quite appealing to the lightweight onboard hardware design for miniaturized satellites, we try to develop a radically new HCS method using just a deterministic addition operator without RPs. We also show that even just with such an addition operator for encoding the information contained in HSI, the proposed all-addition encoder achieves very low sampling rates (e.g., 1%) and, meanwhile, retains sufficient information for the decoder to reconstruct HSI from the noise-corrupted compressed acquisitions. As indicated by several spatial/spectral/global reconstruction quality indices, the proposed AAHCSD does outperform the decoders of other state-of-the-art HCS methods. The mathematical challenge induced by the all-addition scheme is resolved by introducing image self-similarity, a strong prior underlying a series of impressive state-of-the-art contributions to imaging inverse problems, into our convex decoding criterion. Self-similarity has recently been explicitly defined as a convex function. Such explicit form ensures algorithmic convergence, while the convexity enables ADMM-based efficient closed-form solutions, where all the huge dimensional matrix inversions involved in the decoder are judiciously reformulated into equivalent low-complexity ones. Theoretical analysis has also been conducted based on both the conventional BP and recently proposed JE frameworks, where the fundamental role of the latter for hyperspectral analysis has been revealed in recent mathematics and machine learning literature. Experimental evidence clearly shows the superiority of AAHCSD, especially at very low sampling rates. Finally, even though real-time decoding is not necessary as it is performed on the ground station, investigating a real-time solver for the proximal operator associated with the convex self-similarity regularization function is an interesting future line, as such proximal operator should have great potential to be applied to other ill-posed imaging inverse problems.

Appendix

A. Proof of Theorem 1

Based on a more general result by Eckstein and Bertsekas [43, Th. 8], the convergence of AAHCSD has been

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proven in [44] under the following mild assumptions, all satisfied in our case: 1) the terms of the objective function are closed, proper, and convex; 2) the optimization has a solution; 3) the null space of the linear map between the original variables (i.e., $\boldsymbol{s}$ in our case) and the constraint variables (i.e., $(\boldsymbol{u}_1, \boldsymbol{u}_2, \{\boldsymbol{U}_{i,j}\})$ in our case) is just the zero vector; and 4) $\mu > 0$.

Assumption 1) directly follows from the premise of $\alpha_{i,j} \geq 0\ \forall (i,j) \in \mathcal{G} \subseteq \{(i,j) \mid i,j \in \mathcal{I}_{L'},\ i \neq j\}$ and the fact that the composition of a convex function (i.e., $\|\cdot\|_F^2$) and an affine function (i.e., $(\boldsymbol{P}_i - \boldsymbol{P}_j)\boldsymbol{U}_{i,j}$) preserves convexity [26]. Assumptions 2) and 4) are obviously satisfied. As for assumption 3), we recall that the basis matrix $\boldsymbol{E}$ is semiunitary (and hence has full column rank), which, together with some basic property of Kronecker product (i.e., $\text{rank}(\boldsymbol{A} \otimes \boldsymbol{B}) = \text{rank}(\boldsymbol{A})\text{rank}(\boldsymbol{B})$), implies that $\boldsymbol{\Psi}$ is also of full column rank; therefore, when all the constraint variables are zeros, $\boldsymbol{u}_1 = \boldsymbol{\Psi}\widetilde{\boldsymbol{s}}$ is also zero, implying that the coefficient vector $\widetilde{\boldsymbol{s}} = \boldsymbol{0}$. Thus, all the assumptions hold true, and Theorem 1 has been proven. ■

B. Complexity Analysis of Algorithm 1

Due to space limitation, we just show the complexities below without full derivation details. We need a simple fact about the Kronecker product.

Fact 1: Given a matrix $\boldsymbol{M} \in \mathbb{R}^{a\times b}$ and a vector $\boldsymbol{v} \in \mathbb{R}^c$, computing $(\boldsymbol{I}_n \otimes \boldsymbol{M})\boldsymbol{v}$ has a complexity of $\mathcal{O}(nab)$. □

The computational burden of Algorithm 1 lies within the iterative updates of $\boldsymbol{u}_1^{k+1}$, $\boldsymbol{u}_2^{k+1}$, $\{\boldsymbol{U}_{i,j}^{k+1}\}$, and $\boldsymbol{s}^{k+1}$, whose complexities are analyzed as follows.

1) For $\boldsymbol{u}_1^{k+1}$, computing $(\boldsymbol{D}^T\boldsymbol{D} + \mu\boldsymbol{I}_M)^{-1}$ has a complexity of $\mathcal{O}(M^2K + M^3)$. By Fact 1, computing $[(\boldsymbol{I}_L \otimes \boldsymbol{D}^T)\overline{\boldsymbol{y}} + \mu\boldsymbol{v}_1]$ has a complexity of $\mathcal{O}(LKM)$. By Fact 1 again, the matrix–vector multiplication in $\boldsymbol{u}_1^{k+1}$ has a complexity of $\mathcal{O}(LM^2)$. Therefore, computing $\boldsymbol{u}_1^{k+1}$ has a complexity of $\mathcal{O}(M^2K + M^3 + LKM + LM^2)$. If $L \geq M \geq K$, the overall complexity becomes $\mathcal{O}(\zeta M^2(M+L))$, where ζ is the number of algorithmic iterations.
2) For $\boldsymbol{u}_2^{k+1}$, the key observations are that $\boldsymbol{u}_2^{k+1}$ is the vectorization of $\text{vec}_{M\times L}^{-1}[(\boldsymbol{B} \otimes \boldsymbol{I}_M)\underline{\boldsymbol{y}} + \mu\boldsymbol{v}_2]$ $(\boldsymbol{B}\boldsymbol{B}^T + \mu\boldsymbol{I}_L)^{-T}$, the matrix inversion $(\boldsymbol{B}\boldsymbol{B}^T + \mu\boldsymbol{I}_L)^{-1}$ can be obtained by solving an r-dimensional matrix inversion [see (13)], and that $(\boldsymbol{B} \otimes \boldsymbol{I}_M)\underline{\boldsymbol{y}}$ only needs to be computed *once* with a complexity of $\mathcal{O}(rML_h)$. With these observations, the overall complexity can be verified to be $\mathcal{O}(rML_h + \zeta ML + r^3 + \zeta L_h M r^2) = \mathcal{O}(rML_h + \zeta rML)$ if $r \leq \min\{M, L_h\}$.
3) For $\{\boldsymbol{U}_{i,j}^{k+1}\}$, the update of $\boldsymbol{u}_{i,j}^{(c)}$ (i.e., the cth column of $\boldsymbol{U}_{i,j}^{k+1}$) requires first computing $\boldsymbol{\delta}_{i,j}^{(c)}$ (part of which directly gives $[\boldsymbol{u}_{i,j}^{(c)}]_{\mathcal{I}}^{\star}$), having a complexity of $\mathcal{O}(L)$. To compute $[\boldsymbol{u}_{i,j}^{(c)}]_{\mathcal{I}}^{\star}$, the complexity is dominated by $[[\boldsymbol{P}_i - \boldsymbol{P}_j]_{\mathcal{I}}^T[\boldsymbol{P}_i - \boldsymbol{P}_j]_{\mathcal{I}} + ((\mu\boldsymbol{I}_{|\mathcal{I}|})/\lambda\alpha_{i,j})]^{-1}$, having a complexity of $\mathcal{O}(q^2|\mathcal{I}|^2 + |\mathcal{I}|^3)$. Thus, computing $[\boldsymbol{u}_{i,j}^{(c)}]^{\star}$ has a complexity of $\mathcal{O}(L + q^2|\mathcal{I}|^2 + |\mathcal{I}|^3)$, implying that the complexity for updating $\boldsymbol{U}_{i,j}^{k+1}$ is $\mathcal{O}[\zeta N(L + q^2|\mathcal{I}|^2 + |\mathcal{I}|^3)] = \mathcal{O}[\zeta N(L + 3q^2|\mathcal{I}|^2)]$ (recall that $|\mathcal{I}| \leq 2q^2$).
4) For $\boldsymbol{s}^{k+1}$, Fact 1 implies that computing $\boldsymbol{\Psi}^T\boldsymbol{t}_1'$ and $\boldsymbol{\Psi}^T\boldsymbol{t}_2'$ has a complexity of $\mathcal{O}(LNM)$. Since computing $\sum_{(i,j)\in\mathcal{G}} \boldsymbol{t}_{i,j}$ has a complexity of $\mathcal{O}(|\mathcal{G}|LN)$, computing $(\boldsymbol{\Psi}^T\boldsymbol{t}_1' + \boldsymbol{\Psi}^T\boldsymbol{t}_2' + \sum_{(i,j)\in\mathcal{G}} \boldsymbol{t}_{i,j})$ has a complexity of $\mathcal{O}[LN(M + |\mathcal{G}|)]$. Furthermore, computing $(2\boldsymbol{E}^T\boldsymbol{E} + |\mathcal{G}|\boldsymbol{I}_N)^{-1}$ has a complexity of $\mathcal{O}(MN^2 + N^3)$, which only needs to be computed *once*. Finally, by exploiting $(2\boldsymbol{\Psi}^T\boldsymbol{\Psi} + |\mathcal{G}|\boldsymbol{I}_{NL})^{-1} = \boldsymbol{I}_L \otimes (2\boldsymbol{E}^T\boldsymbol{E} + |\mathcal{G}|\boldsymbol{I}_N)^{-1}$ and Fact 1, computing $\boldsymbol{s}^{k+1}$ has a complexity of $\mathcal{O}[\zeta LN(M+|\mathcal{G}|) + MN^2 + N^3 + \zeta LN^2] = \mathcal{O}[\zeta LN(M + |\mathcal{G}|)]$ if $N \leq M \leq L$.

All in all, assuming that $L \geq M \geq \max\{K, N\}$ and $r \leq \min\{M, L_h\}$, the overall complexity is $\mathcal{O}[\zeta M^2(M+L) + rML_h + \zeta rML + \zeta N(L + 3q^2|\mathcal{I}|^2) + \zeta LN(M + |\mathcal{G}|)] = \mathcal{O}[\zeta(M^2L + NL|\mathcal{G}| + Nq^2|\mathcal{I}|^2)]$. ■

C. Proof of Lemma 3

Based on notations in Definition 1, the sampling matrix $\boldsymbol{A}_i$ is the one left-multiplied to the pixel $\boldsymbol{x}_i$. From the definition of $\boldsymbol{\Phi}$, those left-multiplying $\boldsymbol{A}_i$ all come from the upper-half of $\boldsymbol{\Phi}$, i.e., $\boldsymbol{A}_i = \boldsymbol{D} \triangleq \text{bdiag}(\boldsymbol{1}_{M_1}^T, \ldots, \boldsymbol{1}_{M_K}^T) \in \mathbb{R}^{K\times M}$ for all $i \in \mathcal{I}_L$. Thus, we just need to prove the affine independence of columns in $\boldsymbol{D}\boldsymbol{E}$, but this is clearly true (w.p.1.) by noticing

$$[\boldsymbol{D}\boldsymbol{E}]_{kj} \sim \mathcal{V}_k \quad \forall k = 1, \ldots, K$$

where $\mathcal{V}_k$ is a distribution with the probability density function (pdf) $f_k(x) = (1/(2(M_k-1)!))\sum_{i=0}^{M_k}(-1)^i C_i^{M_k}(x - i)^{M_k-1}\text{sgn}(x-i)$ for $x \in [0, M_k]$ (otherwise, $f_k(x) = 0$), in which $\text{sgn}(x) = -1$ for $x < 0$ and $\text{sgn}(x) = 1$ for $x \geq 0$; the notation C_b^a denotes the number of b-combinations from an a-element set. Therefore, columns of $\boldsymbol{D}\boldsymbol{E} \in \mathbb{R}^{K\times N}$ are independently and identically distributed random vectors, each having its kth entry independently drawn from the continuous distribution $\mathcal{V}_k$ [with continuous and strictly positive pdf on $(0, M_k)$]. In other words, each random vector follows a continuous distribution with support on the interior of the K-dimensional hypercube $[0, M_1] \times \cdots \times [0, M_K]$, implying that the N random vectors are affinely independent w.p.1. as long as the number N of random vectors is no greater than one plus the dimension of this hypercube. ■

D. Proof of Theorem 3

In this section, we show that the JE, indeed, yields exact reconstruction $\boldsymbol{x}$ or, equivalently, $\boldsymbol{X} = \boldsymbol{A}\boldsymbol{S}$.

Step 1: We have assumed that the unobservable $\boldsymbol{X}$ satisfies (A1)–(A3), and this allows us to show that the observable hyperspectral data $\overline{\boldsymbol{X}} = \boldsymbol{X}\boldsymbol{B} = \boldsymbol{A}\overline{\boldsymbol{S}}$ with the effective abundances $\overline{\boldsymbol{S}} = \boldsymbol{S}\boldsymbol{B}$ (see Footnote 1) also satisfies the three assumptions, which are verified as follows.

1) Each observable column of $\overline{\boldsymbol{X}}$ is a linear combination of $r = r_1r_2$ columns of $\boldsymbol{X}$, and such linear combination is actually convex combination with combining coefficients $(1/r, \ldots, 1/r)$ [26], as can be seen by verifying the definition of $\boldsymbol{B}$. Furthermore, since $\boldsymbol{X} = \boldsymbol{A}\boldsymbol{S}$ satisfies (A1), we notice that columns (i.e., pixels) in $\boldsymbol{X}$ are all in the convex hull of the columns of $\boldsymbol{A}$, implying

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that the resulting columns in $\overline{\boldsymbol{X}}$ are still in such convex hull conv($\boldsymbol{A}$). This observation, together with $\overline{\boldsymbol{X}} = \boldsymbol{A}\overline{\boldsymbol{S}}$, implies that each column of $\overline{\boldsymbol{S}}$ forms convex combining coefficients, i.e, satisfying the nonnegativity and sum-to-one property; in other words, $\overline{\boldsymbol{X}}$ satisfies (A1).

2) By $\overline{\boldsymbol{X}} = \boldsymbol{A}\overline{\boldsymbol{S}}$, the signatures $\boldsymbol{A}$ remains the same in the observable $\overline{\boldsymbol{X}}$. Thus, $\boldsymbol{X}$ satisfying (A2) directly implies that $\overline{\boldsymbol{X}}$ also satisfies (A2).
3) Since $\overline{\boldsymbol{S}} = \boldsymbol{S}\boldsymbol{B}$ corresponds to an observable γ-pure measurement $\overline{\boldsymbol{X}}$ with $\gamma > 1/\sqrt{N-1}$ (see Footnote 1), conv$\{\overline{\boldsymbol{S}}\}$ must contain $\boldsymbol{f}_i \triangleq (1/(N-1))\sum_{j\neq i}\boldsymbol{e}_j^{(N)}\ \forall i \in \mathcal{I}_N$, as can be directly inferred from the definition of γ. This implies that the column rank of $\overline{\boldsymbol{S}}$ is no less than that of $[\boldsymbol{f}_1, \ldots, \boldsymbol{f}_N]$. As the rank of $[\boldsymbol{f}_1, \ldots, \boldsymbol{f}_N]$ is clearly N, the column rank of $\overline{\boldsymbol{S}}$ is no less N. Thus, the row rank of $\overline{\boldsymbol{S}}$ is no less N and, hence, must be N; recall that $\overline{\boldsymbol{S}}$ has just N rows. Therefore, $\overline{\boldsymbol{S}}$ has full row rank, and $\overline{\boldsymbol{X}}$ satisfies (A3).

Therefore, (A1)–(A3) are satisfied by the observable $\overline{\boldsymbol{X}}$, allowing us to adopt some elegant theory in [23].

Step 2: As the JE criterion is with respect to the observable quantity $\overline{\boldsymbol{X}}$, whose JE (i.e., maximum volume inscribed ellipsoid) $\mathcal{E}^\star \triangleq \mathcal{E}(\boldsymbol{F}^\star, \boldsymbol{c}^\star)$ can be computed from a convex optimization problem (5). Let rbd $\mathcal{X}$ be the relative boundary of the convex hull of column vectors in the observable $\overline{\boldsymbol{X}}$ [45]. Then, by following [23, Th. 3], the contact points $\{\boldsymbol{q}_1, \ldots, \boldsymbol{q}_N\} \triangleq \mathcal{E}^\star \cap (\text{rbd}\ \mathcal{X})$ are also available, from which one can perfectly recover $\boldsymbol{A}$, as illustrated in the following.

Step 3: Results established in Step 1, as well as the premise of $\gamma > 1/\sqrt{N-1}$, tell us that all the assumptions required in proving [23, Corollary 1] are satisfied by the observable HSI $\overline{\boldsymbol{X}}$. Hence, one can exploit [23, Corollary 1] to infer that the JE $\mathcal{E}^\star$ of $\overline{\boldsymbol{X}}$ consequently yields exact reconstruction of $\boldsymbol{A}$ via $\boldsymbol{A} = (\boldsymbol{q}_1 + \cdots + \boldsymbol{q}_N)\boldsymbol{1}_N^T - (N-1)[\boldsymbol{q}_1, \ldots, \boldsymbol{q}_N]$, where $\boldsymbol{q}_i$'s are available as indicated by Step 2. Furthermore, since $\boldsymbol{D}$ is from the measurement design, the quantity $\boldsymbol{D}\boldsymbol{E}$ is also available.

Step 4: Based on notations in Definition 1, the sampling matrix $\boldsymbol{A}_i$ is $\boldsymbol{D}$ for all $i \in \mathcal{I}_L$ (see Appendix C). From the premises of AIP, we know that $\boldsymbol{D}\boldsymbol{E}$ has affinely independent columns. Since $\boldsymbol{\Phi}$ satisfies AIP, Lemma 1 tells us that $\boldsymbol{S}$ can be recovered from $\boldsymbol{A}$ and the HCS measurements $\boldsymbol{A}_1\boldsymbol{x}_1, \ldots, \boldsymbol{A}_L\boldsymbol{x}_L$. Since Step 3 already tells us that $\boldsymbol{A}$ can be reconstructed, $\boldsymbol{S}$ can also be reconstructed if $\boldsymbol{A}_1\boldsymbol{x}_1, \ldots, \boldsymbol{A}_L\boldsymbol{x}_L$ are available. However, $\boldsymbol{A}_1\boldsymbol{x}_1, \ldots, \boldsymbol{A}_L\boldsymbol{x}_L$ are nothing but the upper half of the measurements; specifically, we have $\overline{\boldsymbol{\Phi}}\boldsymbol{x} = (\boldsymbol{I}_L \otimes \boldsymbol{D})\boldsymbol{x}$ that can be further matricized as $[\boldsymbol{D}\boldsymbol{x}_1, \ldots, \boldsymbol{D}\boldsymbol{x}_L]$. Therefore, $\boldsymbol{S}$ is reconstructible, which, together with the JE-reconstructed $\boldsymbol{A}$ (see Step 3), yields the exact reconstruction of $\boldsymbol{X} = \boldsymbol{A}\boldsymbol{S}$. In other words, the JE solution $\widehat{\boldsymbol{x}} \equiv \text{vec}(\boldsymbol{X})$ is exact. ■

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